



BANARAS HINDU UNIVERSITY

B.Sc.(Hons.) Courses

Effective from 2024 under NEP

Offered by

Faculty of Science



BANARAS HINDU UNIVERSITY



Department of Chemistry

Faculty of Science

SYLLABI

B.Sc. (Hons.) Programme

In

Chemistry

Framework of Syllabi for UG according to NEP

Semester	Number of Credits (for Major Course Only)		
	Theory (1 credit- 1 hr/week)	Practicals (1 credit= 2hr/week)	Total
Certificate			
I	3 Credits	1	4
II	3 Credits	1	4
Diploma			
III	Inorganic-I, Organic-I, Physical-I- 2 Credits each	2	8
IV	Inorganic-II, Organic-II, Physical-II, 3 Credits each Techniques in Chemistry- 1 Credit	4	14
B.Sc.			
V	Analytical-I, Inorganic-III- 2 Credits Organic-III, Physical-III- 3 Credits	2+2+2	16
VI	Analytical-II, Inorganic-IV- 3 Credits Organic-IV, Physical-IV- 2 Credits	2+2+2	16
B.Sc. (Hons.)			
VII (without Research)	Analytical-III, Inorganic-V, Organic-V, Physical-V Materials Chemistry- 2 Credits	2+2+2	16
VII (with Research)	Analytical-III, Inorganic-V, Organic-V, Physical-V Materials Chemistry- 2 Credits	2+2+2	
VIII (without Research)	Analytical-IV, Inorganic-VI, Organic-VI, Physical-VI Advanced Spectroscopy- 2 Credits each	2+2+2	16
VIII (with Research)	Advanced Spectroscopy and any one of the following courses: Analytical-IV, Inorganic-VI, Organic-VI, Physical-VI - 2 Credits each	12 (project)	16

CHEMJ11: Basic Concepts of Chemistry-I	
Objectives: To understand the wave nature of matter and its implications. To study the classification of elements and their properties. To study the states of matter; behavior of ideal and real gases, liquids, and the structures of solids. To understand the laws of thermodynamics.	
1. Atomic Structure: Wave-particle duality, Heisenberg uncertainty principle, Schrodinger wave equation, Eigen values and Eigen functions, significance of Ψ and Ψ^2 Radial and angular wave functions, Multi-electron system: Concept of Effective Nuclear Charge (Slater's rules).	5
2. Atomic properties and their periodic trends: A brief introduction to the modern periodic table (Block wise). Atomic size (Covalent, Ionic and van der Waal's radii; Lanthanide contraction), Electron loss, and electron gain enthalpies. Inert pair effect. Electronegativity (Pauling, Mulliken, Allred Rochow's methods of determination): As a criterion of bond type. Lattice energy and its determination: a) Born-Haber cycle b) Born Lande equation c) Comparison of experimental and theoretical values. Solvation/Hydration energy (Solubility of ionic solids in water). Covalency in ionic compounds (Fajan's rules). Redox reaction, Standard electrode potential and its determination. Electrochemical series and its applications Latimer and Frost diagrams; specific illustrative examples.	10
3. Gaseous State: Kinetic theory of gases and ideal gas laws, Maxwell distribution of speed, Equipartition of Energy collisions in a gas: collision number, collision diameter and mean free path, behaviour of real gases: van der Waals equation and its applicability, the critical state: critical constants, law of corresponding states.	10
4. Liquid State: Surface tension, capillary action, measurement of surface tension, viscosity of liquids and its measurement, temperature effect on surface tension and viscosity	4
5. Solid State: Crystal lattices, space lattice, unit cell, crystal systems, law of rational indices, Miller indices, Bragg's equation, crystal structure of NaCl, graphite and diamond	4
6. Chemical Thermodynamics and its Applications: First law of Thermodynamics, Thermodynamic quantities (w, q, ΔU and ΔH) for isothermal and adiabatic reversible expansion of ideal gases, Kirchoff's relations, Joule effect, Joule-Thomson effect, Joule-Thomson coefficient of real (van der Waals) gases. Second Law of Thermodynamics, Carnot cycle, concept of entropy and its physical significance, entropy changes for an ideal gas, entropy changes at constant pressure and volume, entropy change in spontaneous processes, free energy and maximum work functions, conditions for spontaneous changes and equilibrium, Gibbs-Helmholtz equation, Maxwell relations. Statement of third law of Thermodynamics.	12
Outcomes: Students are expected to be able to understand the wave-particle duality, quantum numbers, and screening effect of electrons and its consequences. Students will be able to understand the basis of classification in the Modern Periodic Table and also will be able to explain the properties of various elements. The students will be able to understand the properties of solids, liquids, real and ideal gases. The students will learn various laws of thermodynamics.	
Suggested Books:	
1. Basic Inorganic Chemistry, F. A Cotton, G. Wilkinson, and Paul L. Gaus, 3rd Edition (Indian Adaptation 2021), Wiley. 2. Concise Inorganic Chemistry, J. D. Lee, 5 th Edition (1996), Chapman & Hall, London. 3. Principle of Physical Chemistry, B. R. Puri, L. R. Sharma and M. S. Pathania, Vishal Publishing Co., New Delhi. 4. Physical Chemistry, P. C. Rakshit, 5 th Edition (1988), 4th Reprint (1997), Sarat Book House, Calcutta. 5. Physical Chemistry, P. W. Atkins, 8 th Ed., Oxford University Press, New York. 6. Textbook of Physical Chemistry, K.L. Kapoor, vol. 1, Macmillan India Ltd, New Delhi. 7. Physical Chemistry, G. W. Castellan, 3 rd Ed (1983), Addison-Wesley Publishing Company, Massachusetts.	

CHEMJ12: Lab Work-I	
1. Redox titrations: (a) Redox Titrimetric Estimations Based on Iodometry (b) Redox Titrimetric Estimations with standard $K_2Cr_2O_7$ solution	
2. (a) Determination of Surface Tension of liquids. (b) Determination of Viscosity Coefficients of liquids.	
or	
Inorganic/Organic Chemistry Laboratory: 1. Preparation of molecular models using wire and beads to study (a) Various hybridization geometries (b) Isomerism 2. Identification of Metal ions by borax bead test 3. Preparation of some simple and double salts. 4. Study of displacement of ions in aqueous solutions (e.g. Cu^{2+} by Zn) 5. Preparation of m-dinitrobenzene from nitrobenzene 6. Preparation of acetanilide from aniline	
Note: Experiments may be added/ modified subject to availability of facilities.	
CHEMJ21: Basic Concepts of Chemistry-II	
Objectives: To understand the nature of chemical bonds and its various types using different approaches. To impart knowledge of general reactions mechanisms, preparation and reactivity patterns of alkyl halides, alcohols, ethers, and active methylene compounds.	
1. Chemical Bonding: i) A comparative account of the Valence Bond (VB) and Molecular Orbital (MO) approach of H_2 molecule. Molecular orbital treatment of Homonuclear diatomic molecules of elements of first and second period and Heteronuclear (CO, NO) diatomic molecules. Illustration of HOMO and LUMO. ii) Concept of Hybridization, Bent's Rule, VSEPR theory applied to the compounds of s and p block elements. iii) An introduction to multi-center bonding with illustrative examples of Li-alkyls and diborane. iv) Concept of multiple bonding ($p-p$ and $p-d$) with illustrative examples of BF_3 , P_4O_{10} , $(Me_3Si)_3N$ and SiF_4	6
2. Ionic Bonding: A brief introduction to Crystal Packing, Radius Ratio rule, and its exceptions Structures of a few representative crystal types (Rock salt, Zinc blende, Wurtzite, Rutile, Brookite and Anatase).	5
3. Metallic Bonding: Theories of bonding in metal: Valence Bond, Free electron, and Band theories	2
4. Weak Interactions: Brief introduction to Hydrogen bonding and other weak interactions: Effects on physical and chemical properties.	2
5. Electronic and Field Effect: Electronic (electromeric, inductive, resonance, mesomeric effect, and hyperconjugation) and steric effects on the acidity/basicity of organic molecules.	4
6. Stereochemistry: Fischer, Saw-horse and Newman projection formulae. Chirality-optical activity, enantiomerism and diastereoisomerism involving one and two chiral centers. Configuration: D/L, erythrose/threose, R/S nomenclatures; geometrical isomerism - E/Z nomenclatures. Conformations of n-butane.	5
7. Hydrocarbons: i. Alkenes: Addition reactions (electrophilic and free radical), hydration, hydroxylation, hydroboration, epoxidation, ozonolysis and polymerization. Addition reactions of conjugated dienes and Diels-Alder reaction. ii. Alkynes: Reduction, electrophilic addition, acidity and metal acetylides.	6
8. Alkyl Halides: Nucleophilic substitution: SN_1 , SN_2 and SN_i , Eliminations reactions: E_1 , E_2 , and E_{1cB} mechanisms. Elimination versus substitution reactions. Preparation and synthetic applications of Grignard reagents.	6
9. Alcohols and Ethers: Preparation and reactions of primary, secondary, and tertiary alcohols and ethers.	4
10. Active methylene compounds: Preparation and synthetic applications of ethyl acetoacetate and diethyl malonate. (L)	5

Outcome:	
The students will be able to understand the principles of various types of chemical bonding.	
They will understand the chemistry of hydrocarbons and their derivatives.	
The students will learn the mechanisms of substitution and elimination reactions.	
Suggested Books:	
1. Basic Inorganic Chemistry, F. A Cotton, G. Wilkinson, and Paul L. Gaus, 3rd Edition (Indian Adaptation 2021), Wiley. 2. Concise Inorganic Chemistry, J. D. Lee, 5th Edition (1996), Chapman & Hall, London. 3. Organic Chemistry, I. L. Finar, [Vol. I, 6th Edition (1973), Reprinted in 1980 & Vol. II, 5th Edition (1975), Reprinted in 1996], ELBS and Longman Ltd., New Delhi. 4. Organic Chemistry, 9th Edition, L. G. Wade, J. W. Simek & M. S. Singh, Pearson India Education Services Pvt. Ltd. 2020 (ISBN 978-93-893-4267-3). 5. Organic Chemistry, R. T. Morrison and R. N. Boyd, 6th Edition (1992), Prentice-Hall of India (P) Ltd., New Delhi. 6. Organic Chemistry, Paula Y. Bruice, 2nd Edition, (1998) Prentice-Hall, International Edition.	

CHEMJ22: Lab Work-II	
1. Redox titrations:	
(a) Redox Titrimetric Estimations Based on Iodometry	
(b) Redox Titrimetric Estimations with standard $K_2Cr_2O_7$ solution	
2. (a) Determination of Surface Tension of liquids.	
(b) Determination of Viscosity Coefficients of liquids.	
or	
Inorganic/Organic Chemistry Laboratory:	
1. Preparation of molecular models using wire and beads to study (a) Various hybridization geometries (b) Isomerism 2. Identification of Metal ions by borax bead test 3. Preparation of some simple and double salts. 4. Study of displacement of ions in aqueous solutions (e.g. Cu^{2+} by Zn) 5. Preparation of m-dinitrobenzene from nitrobenzene 6. Preparation of acetanilide from aniline	
Note: Experiments may be added/ modified subject to availability of facilities.	

CHEMJ31: Inorganic Chemistry-I	
Objectives: To study the preparation and chemical nature of some important class of compounds of main group elements. To study industrially relevant inorganic molecules and materials.	
1. Chemistry of main group elements with special references of the following Flame colorations of alkali and alkaline earth metals. Complexes of alkali and alkaline earth metals (the idea of complexation, examples having inorganic, organic chelating ligands, crowns, electrides, and crypts.). Difference between the first and the other members of the group [Be and other elements, B and other elements]. Metalloids. Allotropes (of C, N, P, O, and S) Hydrides, Halides, Hydroxides, Oxo-acids and their salts, and Oxide With special emphasis on (i) Relevant structures (ii) Acidity and Basicity (iii) Redox behaviour (iv) Stabilities (v) Classification and typical examples, and (vi) General methods of preparation of the afore mentioned genres.	18
2. Industrial Chemistry: Silicates with examples of practical utilities (Zeolites. Application in Molecular sieves) Glass, Ceramics, Refractories, Cement, Fertilizers (Nitrogenous and Phosphate fertilizers)	12
Outcomes: The students will understand the structure and properties of some s and p block elements. The students will learn how some industrially important compounds are prepared and how they are important commercially.	
Suggested Books: 1. Basic Inorganic Chemistry, F. A Cotton, G. Wilkinson, and Paul L. Gaus, 3rd Edition (Indian Adaptation 2021), Wiley. 2. Concise Inorganic Chemistry, J. D. Lee, 5th Edition (1996), Chapman & Hall, London. 3. Inorganic Chemistry, Gary Wulfsberg, Iva Student Edition, Reprinted 2014	

CHEMJ32: Organic chemistry-I	
Objectives: To impart a comprehensive understanding of the chemistry involving carbonyl compounds, ylides, aromaticity, their reactivity patterns, and notable reactions in organic chemistry.	
1. Chemistry of Carbonyl compounds: Preparations and reactions - addition and condensation reactions, Cannizzaro, Perkin, Aldol, Benzoin, Haloform, Mannich, and Baylis-Hillman reaction. Important reactions of acids, HVZ reaction, relative reactivity of acid chlorides, acid anhydrides, amides, and esters.	8
2. Chemistry of Phosphorous and Sulphur Ylides: Preparation and Reactions. Wittig reaction, Arbuzov rearrangement, Horner-Woodsworth Emons Modification	5
3. Aromaticity: A general concept of aromaticity antiaromaticity and homoaromaticity. Molecular orbital picture of benzene. Benzenoid and non-benzenoid systems, alternant and nonalternant hydrocarbons.	7
4. Aromatic Electrophilic Substitution: Mechanism of nitration, halogenation, sulphonation, and Friedel-Crafts (alkylation and acylation) reactions. Effects of substituents on <i>ortho/para</i> ratio, orientation and reactivity.	6
5. Preparation and Reactions of Phenols.	4
Outcomes: The students will be able to (a) Understand reaction conditions, products formation and mechanisms of some named reactions. (b) Understand about applications of phosphorous and sulphur ylides. (c) Relate and distinguish various forms of aromaticity and their characteristics. (d) comprehend the reactivity tendencies towards electrophilic aromatic substitution reactions and (e) Learn about the basic chemistry of phenols.	

Suggested Books:	
1.	Organic Chemistry, I. L. Finar, Vol. I, 5th Edition (1975), Reprinted in 1996, ELBS and Longman Ltd., New Delhi.
2.	Advanced Organic Chemistry-Reactions and Mechanisms 2e” – M. S. Singh & K. Raghuvanshi, Pearson Education Pvt. Ltd.
3.	Organic Chemistry”, R. T. Morrison and R. N. Boyd, 6th Edition (1992), Prentice- Hall of India (P) Ltd., New Delhi.
4.	A Guide Book to Mechanism in Organic Chemistry, Peter Sykes, 6th Edition (1997), Orient Longman Ltd., New Delhi.
5.	Advanced Organic Chemistry: Reactions, Mechanisms, and Structure, Jerry March J. Wiley and Sons.
6.	Modern Methods of Organic Synthesis, W. Carruthers and I. Coldham, Cambridge University Press, 2004.
7.	J. Clayden, N. Greeves, S. Warren, and P. Wothers, Organic Chemistry, Oxford University Press, 2014

CHEMJ33: Physical Chemistry-I	
Objectives: To impart the knowledge of electrochemistry and electrochemical cells. They will also learn about various types of equilibria.	
1. Electrochemistry: Arrhenius theory of electrolytic dissociation; Conductance of electrolytes in solution; Ionic conductance, transference number, Determination of transport number by moving boundary method. Kohlrausch's law of independent migration of ions, ion conductance and ionic mobility, applications of conductance measurements, conductometric titrations.	6
2. Chemical Cells and Electromotive Force: Types of Electrodes and electrode reactions, Electrode potential, Standard Electrode potentials, Electrochemical cells, EMF and chemical cell reactions. Free energy, enthalpy and entropy changes of cell reactions; Concentration cells with and without liquid junction, liquid junction potential, salt bridges, applications of EMF measurements. Glass electrode and determination of pH of a solution. Potentiometric titrations: acid-base and redox. Buffer Solutions, Acid-base Indicators	10
3. Phase Equilibria: Thermodynamics of phase transition, Clapeyron-Clausius equation, phase rule, phase diagrams of one-component (water) and two components (phenol-water and Pb-Ag) systems, distribution law, application to cases for dissociation and association, determination of equilibrium constant ($K_1 + I_2 = K_3$).	8
4. Chemical Kinetics: Rate laws for zero, first, and second and Pseudo first-order reactions, steady state approximation, temperature dependence of rates of chemical reactions, Collision theory, qualitative concepts of transition state theory.	6
Outcomes: The students will be able to understand the theories of electrochemistry; mobilities of ions and ionic conductance in solution. They will learn about the working principles of electrochemical cells. The students will be able to understand theories of chemical kinetics and phase equilibria.	
Suggested Books:	
1.	Physical Chemistry, P. C. Rakshit, 5th Edition (1988), 4th Reprint (1997), Sarat Book House, Calcutta.
2.	Textbook of Physical Chemistry, K.L. Kapoor, vol. 3 rd (2005), Macmillan India Ltd, New Delhi.
3.	Physical Chemistry, I. Levine, Tata McGraw Hill, 5th Edn., 2007.
4.	Physical Chemistry, P. W. Atkins, 8th Ed., Oxford University Press, New York 5. Principle of Physical Chemistry, B. R. Puri, L. R. Sharma, M. S. Pathania, Vishal Publishing Co., New Delhi.
5.	Principals of Electrochemistry, D.R. Crow, Taylor & Francis, 4th Ed. 1994
6.	Chemical Kinetics, K. J. Laidler, Pearson India, 3 rd Ed.

CHEMJ34: Qualitative Analysis and Thermochemistry	
1(a) Systematic semimicroanalysis of inorganic salts Mixtures containing 2 cations and 2 anions	
1(b). Qualitative analysis of simple organic compounds: Preliminary Test Element detection Functional group Test Melting point measurement	
2. Thermochemistry: Determination of water equivalent of a calorimeter and the enthalpy of neutralization of hydrochloric acid with sodium hydroxide. Determination of enthalpy of ionization of ethanoic acid. Determination of integral enthalpy of solution of salts	

CHEMJ41: Inorganic Chemistry-II	
Objectives: To understand the various theories of acids and bases To study a representative non-aqueous solvent. To study the descriptive chemistry of transition metals, lanthanides and actinides.	
1. Acids and Bases: Brønsted–Lowry, Lux-Flood and Solvent system concepts of acids and bases, leveling effect, Lewis concept of acids and bases and factors affecting their strength, HSAB theory applications: Characteristics of hard-soft acids and bases and borderline cases. Prediction of the feasibility of reactions and stability of compounds, bonding preferences, occurrence of metals as ores in nature, and symbiosis. Super-acids and frustrated acid-base pairs.	9
2. Non-aqueous Solvent: Significance of a solvent towards chemical reactions and limitations of water as a solvent. Classification and solvent physical properties for being an effective reaction medium, viz. melting and boiling points, heat of fusion and vaporization, dielectric constant, dipole moment, and viscosity. Liquid NH ₃ as a nonaqueous solvent, chemical reactions, and alkali metal solutions in liquid NH ₃ . A brief introduction to solvent-free, ionic liquid, and green synthesis.	6
3. Transition Metal Chemistry: Characteristics of <i>d</i> -block elements with special reference to 3d transition series (electronic configuration, oxidation state, color, spin only magnetic moment, complex formation, and catalytic properties).	8
4. Coordination compounds: Molecular addition compounds: double salts and coordination compounds. Definition of ligand, denticity. Werner's theory, nomenclature of coordination complexes. Sidgwick's EAN concept and Valence Bond Theory, outer/inner orbital complexes. Geometrical isomerism of coordination compounds with coordination nos. 4 and 6 (Ma ₆ , Ma ₅ b, Ma ₄ b ₂ , Ma ₃ b ₃ , M(aa) ₃ , M(aa) ₂ b ₂ type). Optical isomerism.	12
5. Lanthanides: Separation of Lanthanide elements (Various methods in brief). Comparative study of lanthanide elements with respect to electronic configuration, atomic and ionic radii, oxidation state, lanthanide contraction, and complex formation tendency.	5
6. Actinides: Occurrence and preparation of actinide elements. Special reference/method to the enrichment of Uranium. Comparative study of actinide elements with respect to electronic configuration, atomic and ionic radii, oxidation states, compounds, and complex formation tendency.	5
Outcomes: The students will learn the theories that explain various types of acid-base behaviour. They will learn the importance of non-aqueous solvents and their applications. The students will be able to understand the bonding in coordination compounds including the characteristics of transition metals, lanthanides, and actinides.	
Suggested Books: 1. Inorganic Chemistry, J. E. Huheey, E.A. Keiter and R.L. Keiter, 4th Edition (2006), Addison Wesley Publishing Co, NY. 2. Basic Inorganic Chemistry, F. A Cotton, G. Wilkinson, and Paul L. Gaus, 3rd Edition (1995), John Wiley & Sons, New York. 3. Concise Inorganic Chemistry, J. D. Lee, 5th Edition (1996), Chapman & Hall, London. 4. Inorganic Chemistry, Gary L. Miessler and Donald A. Tarr, 3rd Edition (2008), Pearson, India Education Service Pvt Ltd.	

CHEMJ42: Organic Chemistry-II	
Objectives: To make the students aware of the chemistry of various classes of organic compounds	
1. Nitrogen containing compounds: Preparation and reactions of nitrobenzene and aniline. Preparation and synthetic applications of Diazonium salts.	5
2. Heterocyclic Compounds: Nomenclature of heterocyclic compounds, synthesis and reactions of furan, pyrrole, thiophene, pyridine, indole and quinoline.	13
3. Polynuclear Hydrocarbons: Chemistry of naphthalene and anthracene.	
3. Colour and Constitution: Synthesis of malachite green, fluorescein, Indigotin.	5
4. Alicyclic compounds: Cycloalkanes, general synthesis, Bayer's strain theory. Cyclohexane: chair and boat conformations, conformation of mono substituted and disubstituted cyclohexanes.	6
5. Protection and Deprotection of Functional Groups: protection of NH group, OH group, diols, carbonyl groups, carboxyl groups, double bonds and triple bonds.	8
6. Carbohydrates: Characteristic reactions of aldoses and ketoses. Glucose-structure (Open and Cyclic), Mutarotation, anomeric effect, iminosugar and sialic acids. Sucrose, starch and cellulose (Structural aspects only), glycosylation reactions.	8
Outcomes: The students will be able to understand a. The fundamental chemistry of diazonium salts, heterocyclic compounds, dyes, alicyclic compounds and carbohydrates. b. Pros and cons of protection and deprotection strategies in organic transformations. Suggested Books: 1. Organic Chemistry", R. T. Morrison and R. N. Boyd, 6th Edition (1992), Prentice-Hall of India (P) Ltd., New Delhi. 2. Organic Chemistry", 9th Edition, L. G. Wade, J. W. Simek & M. S. Singh, Pearson India Education Services Pvt. Ltd. 2020 (ISBN 978-93-893-4267-3) 3. Organic Chemistry, I. L. Finar, [Vol. I, 6th Edition (1973), Reprinted in 1980 & Vol. II, 5th Edition (1975), Reprinted in 1996], ELBS and Longman Ltd., New Delhi. 4. Organic Chemistry, J. Clayden, N. Greeves, S. Warren, and E. Wothers, (2001) Oxford Univ. Press, Oxford. 5. Carbohydrates in drug discovery and development, V. K. Tiwari, Elsevier Inc. 2020. 6. Advanced Organic Chemistry: Reactions, Mechanisms, and Structure, Jerry March J. Wiley and Sons. 7. Heterocyclic Chemistry, Thomas L. Gilchrist; Edition, 5; Publisher, World Publishing Corporation, 1997; ISBN, 7506233983 8. P. G. M. Wuts and T. W. Greene, Greene; Protective Groups in Organic Synthesis, 4th Edition, 2006.	

CHEMJ43: Physical Chemistry-II	
Objectives: To impart knowledge on the application of classical thermodynamics on solution, surface and photochemistry and nuclear chemistry	
1. Thermodynamics of Solutions: Partial molal quantities, chemical potential of a mixture of ideal gases, Gibbs-Duhem equation, chemical potential of real gases and fugacity, activity and activity coefficient, variation of fugacity with temperature and pressure, Lewis-Randall rule, thermodynamic functions of mixing (ΔG_{mix} , ΔS_{mix} , ΔV_{mix} , ΔH_{mix}), ideal solutions and their characteristic properties, Duhem-Margules equation and its applications, Henry and Raoult's laws, thermodynamics of colligative properties, relative lowering of vapour pressure, freezing point depression, elevation of boiling point and osmotic pressure, van't Hoff equation, measurement of osmotic pressure and determination of molecular weight of macromolecules.	15

2. Surface and Photochemistry: Langmuir adsorption isotherm, Multi-layer adsorption-BET equation (no derivation) and its application to surface area measurements; Gibbs Adsorption isotherm, surface reactions: kinetics of unimolecular surface reactions, inhibition and activation energy, nature of surface. Enzyme kinetics (Michaelis-Menten equation), Photochemical reactions: quantum efficiency, reasons for low and high quantum efficiency, kinetics of photochemical reactions: decomposition of HI, H ₂ -Br ₂ reaction, photostationary state: dimerisation of anthracene, Stern-Volmer relation.	15
3. Nuclear Chemistry: Radioactivity: general characteristics of radioactive decay kinetics, artificial radioactivity, detection and measurement of radioactivity: GM counter. Nuclear Reactions: Types of nuclear reactions, conservation of linear momentum and mass- energy in nuclear reactions, nuclear reaction cross section. Compound nucleus theory and its experimental verification. Radio chemistry: Probing by isotopes, preparation of radioisotopes, radiochemical principles in use of tracers, some typical applications: chemical, physicochemical and analytical applications, age determination. Radiation Chemistry: Interaction of nuclear radiations with matter- charged particle, neutrons and gamma rays. Radiolysis of water and aqueous solutions, radiation dosimetry.	15
Outcomes: After the completion of the course, the students should be able to 1. Apply the concept of chemical potential to explain the Gibbs free energy change due to mixing and calculate partial molar quantities. 2. Understand the concept of fugacity, activity, ideal and nonideal solution. 3. Explain the thermodynamic basis of colligative properties and applications 4. Learn about surface phenomenon, adsorption isotherms, BET Equation, the laws of absorption of light energy by molecules and the subsequent photochemical reactions 5. Understand the basics of nuclear chemistry	
Suggested Books: 1. Physical Chemistry, P. C. Rakshit, 5th Edition (1988), Sarat Book House, Calcutta. 2. Textbook of Physical Chemistry, K.L. Kapoor, vol. 5th (2005), Macmillan India Ltd, New Delhi. 3. Essentials of Nuclear Chemistry, H.J. Arnikar, 4th Edition (2003), New Age International Publishers, New Delhi. 4. Physical Chemistry, I. Levine, 5th Ed., (2007), Tata McGraw Hill New Delhi. 5. Physical Chemistry, P. W. Atkins, 8th Ed., Oxford University Press, New York.	

CHEMJ44: Techniques in Chemistry	
Objectives: To introduce basic concepts about stoichiometry, acid-base equilibria, and separation methods.	
1. Introduction to Separation and Instrumental Methods Distillation, Crystallization, Precipitation, Extraction, Chromatography, Introductory ideas of UV-visible spectroscopy, Infrared Spectroscopy, and NMR Spectroscopy.	8
2. General Introduction to Analysis, and Stoichiometric Calculations Qualitative and quantitative analysis, primary standard, secondary standard, preparation of standard solutions, molarity, normality, molality, weight/weight percentage, weight/volume percentage, and ppm.	4
3. Concepts of Acid-Base Equilibria and Titrations General concepts of acids and bases, strength of conjugate acid/base pair, acid-base equilibria in water, pH calculations for strong acids and strong bases, concepts of acid base titration, complexometric titration, redox titration, and precipitation titration, indicators.	3
Outcomes: The student will understand about qualitative and quantitative analysis, titrations, and instrumentations on the separation techniques.	
Suggested Books: 1. Daniel C. Harris, Quantitative Chemical Analysis, 9th Edition (2015), W. H. Freeman and Company, New York. 2. Basic Concepts of Analytical Chemistry', S.M. Khopkar, 2nd edition (1998), New Age International Publications, New Delhi. 3. Analytical Chemistry, G.D. Christian, (2001) John Wiley & sons, New York 4. Fundamentals of Analytical Chemistry, D.A. Skoog, D. M. West, F. James, Cengage Learning India, 10th Ed. 2022	

CHEMJ45: Physical Chemistry Lab Work	
1. Complexometric titrations with EDTA: Mg^{2+} , Zn^{2+} estimation. 2. Determination of the hardness of water. 3. Kinetics study of acid catalysed hydrolysis of methyl acetate. 4. Determination of pH of a given buffer by colour matching method. 5. Determination of Critical Solution Temperature (CST) of Phenol-Water system. 6. Effect of impurity on Critical Solution Temperature of Phenol-Water system	
Note: Experiments may be added/ modified subject to availability of facilities.	

CHEMJ46: Organic, Inorganic Chemistry Lab Work	
A. Preparation of organic compounds: 1. Acetanilide from aniline 2. p-Bromo acetanilide from acetanilide 3. Phthalic anhydride from phthalic acid 4. Preparation of phenyl-azo- β -naphthol (an Azo dye). 5. Preparation of aniline yellow (p-amino-azobenzene). 6. Preparation of Schiff base (Salicylaldehyde and aniline)	
B. Preparation of Some Inorganic complexes (Including those using the prepared Schiffbase as ligand) Purification by recrystallization of the compounds.	
C. Preparation of standard solutions (both primary and secondary) of a particular normality/ molarity.	
D. Application of Separation and Purification Techniques.	
Note: Experiments may be added/ modified subject to availability of facilities.	

CHEMJ51: Analytical Chemistry-I	
Objectives: To learn the basics of chemical analysis, the use of statistical methods, and gravimetric analysis	
1. Statistical Evaluation: Chemical analysis, Classification of analytical methods, Steps in a quantitative analysis, Quantitative range, Determinate and indeterminate errors, Normal error curve, Accuracy and Precision, Relative error, Relative standard deviation, Methods for minimizing errors, Criteria for rejection of observation, Significant figures and computation rules, Error propagation.	8
2. Recent trends for chemical analysis: Method selection, method validation, sample preparation, Analytical validation of results, LOD and LOQ, Good lab practice, Good manufacturing practice.	7
3. Precipitation: Desirable properties of gravimetric precipitates, Formation of gravimetric precipitates, Conditions for quantitative precipitations, Contamination in precipitates, Method for removal of impurities in precipitates, Steps involved in quantitative precipitation, Organic precipitants (oxine, dithizone, α -nitroso-(naphthol, cupferon, dimethyl glyoxime) in chemical analysis.	15
Outcomes: The student will understand about the recent developments in chemical analysis, including gravimetric analysis	
Suggested Books: 1. Modern Methods of Chemical Analysis', R.L. Pecscock, L.D. Shields, T. Cairns, and I.C. Mc William, 2 nd Edition (1976), John Wiley, New York. 2. Basic Concepts of Analytical Chemistry', S.M. Khopkar, 2 nd edition (1998), New Age International Publications, New Delhi. 3. Analytical Chemistry', G.D. Christian, (2001) John Wiley & sons, New York 4. Environmental Chemistry', A.K. De, 3 rd edition (1994), Wiley Eastern, New Delhi.	

CHEMJ52: Inorganic Chemistry-III	
Objectives: To teach the theories of metal-ligand bonding, their drawbacks and applications To impart the knowledge of electronic spectra of transition metal complexes.To impart the knowledge of reactivities of transition metal complexes	
1. Theories of Metal-Ligand bonding: Crystal-field theory and crystal-field splitting in octahedral, tetrahedral, and square planar complexes. Jahn-Teller distortion. Factors affecting the crystal-field splitting, Spinel, Inverse spinel, OSSE, Spectrochemical and Nephelauxetic series. Concept of Metal-ligand orbital overlap: ligand field theory, molecular orbital theory with reference to octahedral 3d transition metal complexes (without-bonding, and with bonding).	14
2. Electronic Spectra of Transition Metal Complexes: Types of electronic transitions; static and dynamic electronic transitions, selection rules for d-d transitions, spectroscopic ground states. Explanation of electronic spectra on the basis of Orgel energy level diagrams for d^1 - d^{10} states. Charge transfer spectra; electronic spectra of octahedral and tetrahedral Co(II) and Ni(II) complexes and calculation of ligand-field parameters.	10
3. Thermodynamic and Kinetic aspects of Metal Complexes: A brief outline of thermodynamic and kinetic stabilities of metal complexes and factors affecting their stability. Substitution reactions of square-planar complexes, Trans effect.	6
Outcomes: The student will be able to understand the nature of metal-ligand bonding in transition metal complexes. They will be able to explain the electronic spectra of the transition metal complexes and also their reactivity.	
Suggested Books: 1. "Concise Inorganic Chemistry", J. D. Lee, 5th Edition (1996), Chapman & Hall, London. 2. "Inorganic Chemistry", J.E. Huheey, E.A. Keiter and R.L. Keiter, 4th Edition (2006), Addison Wesley Publishing Co, NY. 3. "Basic Inorganic Chemistry", F. A Cotton, G. Wilkinson, and Paul L. Gaus, 3rd Edition (1995), John Wiley & Sons, New York. 4. "Inorganic Chemistry", A. G. Sharpe, 3rd International Student Edition (1999), ELBS / Longman, U.K. 5. "Inorganic Chemistry", D. F. Shriver and P. W. Atkins, 3rd Edition (1999), ELBS, London.	

CHEMJ53: Organic Chemistry-III	
Objectives: To impart knowledge of stereochemistry, reactive intermediates, isotope effect, photo chemistry, and notable reactions in organic chemistry.	
1. Stereochemistry: Symmetry elements, symmetry operations, racemates, racemization, resolution (chemical and enzymatic), pro-chirality, concept of homotopic and heterotopic, pro-stereoisomerism with suitable examples of one and two chiral centers. Asymmetric induction, Cram's rule: Addition of nucleophile to carbonyl function. Aldol reaction (achiral-achiral). Regioselective, chemoselective, stereospecific and stereoselective reactions. Estimation of optical purity (ee and de). Axial chirality (e. g., allenes, biphenyls and spiranes).	10
2. Reactive intermediates: Generation, stability and reactivity of carbocations, carbanions, free-radicals, carbenes, and nitrenes.	8
3. Isotope effect: Isotopic substitution in a molecule, primary and secondary kinetic isotope effects, solvent isotopic effect and their importance in mechanistic studies.	5
4. Name reactions: Michael addition, Mitsunobu reaction, Birch reduction, Wagner-Meerwein, Baeyer-Villiger oxidation, Robinson annulation, Pinacol-pinacolone rearrangement, Wolff rearrangement, dienone-phenol rearrangement, benzidine rearrangement, Favorskii rearrangement. Hofmann, Curtius, & Schmidt rearrangements, Click chemistry.	12
5. Photochemistry: Principles of photochemistry, Energy (inter and intramolecular) transfer, Photosensitisation, Singlet and Triplet state photochemical reactions of 1,3-butadiene, cis-trans isomerism, Inter and intramolecular cycloaddition reactions, Di- π -methane rearrangement. Photochemical reaction of carbonyl compounds: Photo reduction, Norrish type-I cleavage of acyclic and cyclic compounds, Norrish type-II cleavage. Photocyclo-addition reaction: Paterno-Buchi reaction. Photo-Fries rearrangement.	10

Outcomes: The students will be able to understand the 1. Basic chemistry of stereochemistry, reactive intermediates, and isotope effect inorganic molecules. 2. Reaction conditions, product formation, and mechanisms of specific rearrangement reactions commonly employed in organic synthesis. 3. Reactions involving photochemical process for organic molecules.	
Suggested Books: 1. Organic Chemistry, I. L. Finar, Vol. I, 5th Edition (1975), Reprinted in 1996, ELBS and Longman Ltd., New Delhi. 2. Advanced Organic Chemistry-Reactions and Mechanisms 2^e—M. S. Singh & K. Raghuvanshi, Pearson Education Pvt. Ltd., 2021. 3. Advanced Organic Chemistry: Reactions, Mechanisms, and Structure, Jerry March J. Wiley and Sons 4. Stereochemistry of Organic Compounds, D. Nasipuri, 2nd Edition (1994), Wiley Eastern Ltd., New Delhi 5. Stereochemistry of Organic Compounds, E. L. Eliel, S. H. Wilen and Mander, (2004) Wiley Interscience, New York. 6. Organic Chemistry, J. Clayden, N. Greeves, S. Warren, and E. Wothers, (2001) Oxford Univ. Press, Oxford. 7. Reactive Intermediates in Organic Chemistry-Structure, Mechanisms, and Reactions, M. S. Singh Wiley-VCH, Germany, (2014)	

CHEMJ54: Physical Chemistry-III
Objectives: To make students understand the limitations of classical mechanics and the necessity of quantum chemistry, introduce them with postulates of quantum chemistry and apply the same to derive equations for various models and hydrogen atoms. Understand the basis of molecular spectroscopy and its applications.
1. Elementary Quantum Mechanics: quantum mechanical operators, Hamiltonian operator, Schrödinger wave equation and its importance, Eigen functions, Eigen values, physical interpretation of the wave function, orthogonality of wave functions, Heisenberg's uncertainty principle, the particle in a one-dimensional box problem and its solutions, particle in a three dimensional box, degeneracy, rigid rotor and harmonic oscillator.
2. Molecular Symmetry: Symmetry elements and operations. Point groups. Matrix representation of symmetry elements and point groups, matrices of C_{3v} and C_{4v} point groups, Structure of character tables, determination of symmetry species for translations and rotations.
3. Molecular Spectroscopy: Region of electromagnetic spectrum, Interaction of electromagnetic radiation with molecules and various types of spectra, signal to noise ratio and resolving power, width and intensity of spectral transitions, Born Oppenheimer approximation, pure rotational spectra, diatomic rigid rotor molecules, effect of isotope substitution, vibrational and vibration-rotational spectra of diatomic molecules, harmonic oscillator-rigid rotor approximation, anharmonicity effect, normal modes of vibration, infrared spectra of linear and bent AB₂ molecules, Franck-Condon principle, electronic spectra of diatomic molecules, vibrational structure. Nuclear Magnetic Resonance Spectroscopy: Introduction, chemical shifts, spin-spin splitting, relaxation times
Outcomes: After the completion of the course, the students should be able to 1. Learn about the limitations of classical mechanics and how quantum mechanics can be used to solve problems with atomic and molecular systems. 2. Gain knowledge of quantum mechanics, quantization, and probability distribution, the uncertainty principle, and spectroscopy & use of quantization. 3. Understand how to interpret distinct spectra and how they might be used to reveal structures.

Suggested Books:

1. Physical Chemistry, P. Atkins and J. De Paul, 8th Edition (2006), International Student Edition, Oxford University Press.
2. Physical Chemistry - A Molecular Approach, D. A. Mc Quarrie and J. D. Simon, Viva Books, New Delhi, 2004.
3. Symmetry and Spectroscopy of Molecules, K. Veera Reddy, Paperback, New Age International Private Limited, 2007.
4. Fundamentals of Molecular Spectroscopy, C. N. Banwell, 4th Edn. (2017), Paperback, McGraw Hill Education (India) Private Limited, New Delhi.
5. Introduction to Molecular Spectroscopy, G.M. Barrow, Paperback, McGraw Hill, 2018.
6. Introductory Quantum Chemistry, A. K. Chandra, 4th Edn. (2017), McGraw Hill Education (India) Private Limited.
7. Chemical Applications of Group Theory, Cotton, F. A., Wiley India, 3rd Ed. 2017
8. Basic Organometallic Chemistry, B.D. Gupta and A.J. Elias, Universities Press, 2nd Ed. 2013

CHEMJ55: Physical Chemistry Lab Work

1. Composition determination of an unknown binary liquid mixture from surface tension-composition curve.
2. Composition determination of an unknown binary liquid mixture from viscosity composition curve.
3. Conductometric titration of strong acid against strong alkali.
4. Conductometric titration of weak acid against strong alkali.
5. Potentiometric titration of strong acid against strong alkali.
6. Study the kinetics of the catalytic decomposition of H₂O₂ by chemical method.
7. Verify Lambert-Beer's law and determine the concentration of an unknown CuSO₄/KMnO₄/K₂Cr₂O₇ solution.

Note: Experiments may be added/ modified subject to availability of facilities.

CHEMJ56: Systematic Organic Analysis

Systematic identification of organic compounds and derivatization

Note: Experiments may be added/ modified subject to availability of facilities.

CHEMJ61: Analytical Chemistry-II**Objectives:**

To gain knowledge on advanced separation techniques, analytical reagents, and UV-Vis molecular absorbance spectroscopy

1. Solvent Extraction and Chromatography: Distribution law, Single extraction, Multiple extraction, Craig concept of counter-current distribution, Important solvent systems: chelate extraction, synergic extraction, extraction by solvation, ion-pair extraction, Basics and classification of chromatographic methods, paper chromatography, adsorption chromatography, and ion-exchange chromatography. **15**

2. Analytical Reagents: Theoretical and practical aspects of the use of EDTA, cerate, iodate, bromate, chloramine-T, Karl Fischer and Malaprade (periodate) reagents in chemical analysis. **15**

3. Spectrophotometry: Beer's law and application, Nomenclature and units, General instrumentations for spectrophotometry, Spectrophotometric determinations of one component (iron, chromium, and manganese) and two components (overlapping and non-overlapping) systems, Spectrophotometric determination of dissociation constants of indicator, Photometric errors. **15**

Outcomes:

The students will understand the widely used techniques like, chromatography, analytical reagents and spectrophotometry.

Suggested Books:	
1.	Principles of Instrumental Analysis, D.A. Skoog, F.J.Holler and T.A. Nieman, 5 th edition (1998), Horcourt Brace & Company, Florida.
2.	Analytical Chemistry, G.D. Christian, (2001) John Willey & sons, New York.
3.	Instrumental Methods of Analysis', H.H. Willard, L.L. Merritt, and J.A. Dean, 6 th Edition (1986), CBS Publishers & Distributors, Shahdara, Delhi.
4.	Modern Methods of Chemical Analysis, R.L. Peacock, L.D. Shields, T. Cairns, and I.C. Mc William, 2 nd Edition (1976), John Willey, New York.

CHEMJ62: Inorganic Chemistry-IV	
Objectives: To provide standard guidance to students about the magnetic properties of transition metal chemistry and their anomalies. To give a comparative approach among the chemistry of metals belonging to 3 rd , 4 th and 5 th transition series of elements and their uses. To train students about the organometallic synthesis using industrially important metals. To educate the students about some functional inorganic materials.	
1. Magnetic Properties of Transition Metal Complexes: Types of magnetic behavior, methods of determining magnetic susceptibility, <i>L-S</i> and <i>J-J</i> coupling, orbital contribution to magnetic moments. Correlation of magnetic moment data and stereochemistry of Co(II) and Ni(II) complexes; anomalous magnetic moments.	10
2. Chemistry of Second and Third Transition Series: A general comparative treatment of 4d and 5d elements with their 3d analogs in respect of ionic radii, oxidation states, complex formation tendency, magnetic behavior, and electronic spectral properties.	8
3. Organometallic Chemistry: Definition, nomenclature, and classification of organometallic compounds. Preparation, properties, bonding, and applications of alkyls and aryls of Li, Al, Sn, and Ti. Preparation, structure, and properties. Bonding in metal carbonyls, variants of CO bridging, vibrational spectra of metal Carbonyls. A brief account of metal-ethylenic complexes. Concept of catalyst designing. Homogeneous hydrogenation (Wilkinson's catalyst) Heterogeneous (Ziegler-Natta catalyst) catalyst, a brief introduction to catalytic cycle.	17
4. Functional materials: Structural and applicative aspects in brief: Diamond, Graphite, Graphene, CNT, Graphyne, Quantum dots, Fullerene (C60). Intercalation compounds and Clathrates. Soft Materials	10
Outcomes: Students shall be able to	
1. Learn the magnetic properties of transition metals and related compounds as well as their anomalous behavior.	
2. Significantly understand the concept of metal-to-carbon bonding and its applications.	
3. Understand the need and importance of functional materials as well as their synthesis and properties.	
Suggested Books:	
1.	Concise Inorganic Chemistry, J. D. Lee, 5th Edition (1996), Chapman & Hall, London.
2.	Inorganic Chemistry, J. E. Huheey, E.A. Keiter and R.L. Keiter, 4th Edition (2006), Addison Wesley Publishing Co, NY.
3.	Basic Inorganic Chemistry, F.A. Cotton, G. Wilkinson, and Paul L. Gaus, 3rd Edition (1995), John Wiley & Sons, New York.
4.	Inorganic Chemistry A. G. Sharpe, 3rd International Student Edition (1999), ELBS/Longman, U.K.
5.	Elements of Magneto chemistry, R.L. Dutta and A. Shyamal, (1993) Affiliated EastWest Press Ltd
6.	Organometallic Chemistry, R. C. Mehrotra and A. Singh
7.	Modern Inorganic Chemistry, William Jolly, Tata McGraw Hill 2nd Ed.

CHEMJ63: Organic Chemistry-IV	
Objectives: To impart knowledge of amino acids, proteins, polymers, terpenes, vitamins, and hormones.	
1. Amino acids and Peptides: Amino acids - preparations, physical properties, dipolar nature, isoelectric point, chemical reactions, and configuration. Peptides: peptide linkage, peptide synthesis and structure of poly peptides, C and N terminal amino acid determination. Proteins: General characteristics and	10

secondary structure.	
2. Polymers: Types of polymers and polymerization process: Addition (radical, ionic), step growth (condensation), and stereo-controlled (coordination) polymerization mechanisms. Synthesis and uses of following polymers: (i) Plastics: PVC, PE, and Teflon, (ii) Specialty polymers: Conducting & electroluminescent (Organic light emitting diodes), Liquid crystals (iii) Natural and synthetic rubber (iv) Fibers: Rayon, Polyester, Polyamides (v) Foaming agent and plasticizers (vi) Biodegradable polymers.	8
3. Terpenes: Occurrence, isolation and classification. Isoprene rule of terpenes. Structure and synthesis of Citral, α -terpinol, and camphor.	7
4. Vitamin and hormones: Chemical constitution and synthesis of vitamin A, vitamin C and thyroxine.	5
Outcomes: The students will be able to understand 1. The structure and characteristics of amino acids, peptides, proteins and carbohydrates. 2. the structure elucidation for some natural products isolated from terpenes and alkaloids. 3. the fundamental concepts of polymer synthesis and its various properties.	
Suggested Books: 1. Organic Chemistry, I. L. Finar, Vol. I, 5th Edition (1975) Reprinted in 1996, ELBS and Longman Ltd, New Delhi. 2. Organic Chemistry, I. L. Finar, Vol. II, 5th Edition (1975) Reprinted in 1996, ELBS and Longman Ltd, New Delhi. 3. Organic Polymer Chemistry, K. J. Saunders, 2nd Edition (1988), Chapman & Hall, London. 4. Introduction to Synthetic Polymers, Ian M. Campbell, 2nd Edition (2000), Oxford University Press, USA. 5. Principles of Polymer Science, P. Bahadur and N. V. Sastry, (2002) Narosa, New Delhi.	

CHEMJ64: Physical Chemistry-IV	
Objectives: To impart knowledge on different type of interaction in electrolytic solution and Debye Huckel theory, statistical concept of thermodynamics and kinetics of relatively complex reactions.	
1. Electrochemistry: Theory of strong electrolytes: qualitative idea of Debye-Hückel theory of ion-ion interactions, Debye-Hückel limiting law for activity coefficient of ions in electrolyte solution (derivation not required), its modification for concentrated solutions, Debye-Hückel-Onsager (D-H-O) theory of electrolytic conductance: qualitative idea of electrophoretic and relaxation effects, D-H-O equation for conductance of electrolyte solutions. Effect of high frequency and high field on conductance.	7
2. Advanced Chemical Kinetics: Kinetics of complex reactions (reversible, simultaneous, and consecutive), chain reactions; dynamic chain (Hydrogen-Bromine reaction, decomposition of ethane), branched-chain (Hydrogen-Oxygen reaction), oscillatory reactions (Belousov-Zhabotinskii reaction), experimental techniques for fast reactions (stopped-flow, temperature- jump and flash photolysis).	8
3. Molecular Statistics: The Boltzmann distribution, Maxwell-Boltzmann distribution law for the distribution of molecular energies, partition functions, thermodynamic quantities from partition functions, Sackur-Tetrode equation for molar entropy of monatomic gases, rotational and vibrational partition functions, calculation of Gibbs free energy changes and equilibrium constant in terms of partition functions.	15
Outcomes: After the completion of the course, the students should be able to 1. Describe the activity coefficients of electrolyte solution as a function of concentration. 2. Understand ionic atmosphere and the dependence of conductance on concentration 3. Demonstrate the relation between microscopic and macroscopic properties of matter. 4. Understand the kinetics and rate equation of some relatively complex reactions	
Suggested Books: 1. Modern Electrochemistry, vol. 2 A & B, J.O'M. Bockris and A. K. N. Reddy, Plenum Press, New York. 2. Chemical Kinetics, K. J. Laidler, Harper & Row, New York. 3. Physical Chemistry, P. Atkins and J. De Paul, 8th Edition (2006), International Student Edition, Oxford University Press.	

CHEMJ65: Physical Chemistry Lab Work	
1. Order of reaction of I ₂ / Acetone / H ⁺ .	
2. Study of the kinetics of acid-catalyzed hydrolysis of sugar by chemical method.	
3. Conductometric titration of KCl with AgNO ₃ .	
4. Conductometric titration of a mixture of weak and strong acids.	
5. Distribution coefficient of acetic acid between <i>n</i> -butanol and water solvents.	
6. Potentiometric titration of a redox system (ferrous ammonium sulfate with K ₂ Cr ₂ O ₇).	
7. Determination of relative strengths of two acids by studying the kinetics of acid-catalyzed ester hydrolysis.	
8. Determination of relative strengths of two FeCl ₃ solutions by studying the kinetics of FeCl ₃ -catalysed H ₂ O ₂ decomposition gasometrically.	
9. Determination of isosbestic point and indicator constant	

CHEMJ66: Quantitative Inorganic Analysis	
Quantitative estimation of Metal ions by Gravimetric method (Fe ³⁺ , Ba ²⁺ , Ni ²⁺ , Co ²⁺ , Hg ²⁺) Separation and identification of metal ions by paper chromatography	

CHEMJ67: Organic Chemistry Lab Work	
Preparation of derivatives: 1. Preparation of SBT from thiourea 2. Preparation of Phthalimide from phthalic anhydride 3. Preparation of dibenzylidene acetone from benzaldehyde 4. Preparation of hippuric acid from glycine 5. Preparation of benzyl from benzoin 6. Preparation of Benzilic acid from benzyl	

CHEMJ71: Analytical Chemistry-III	
Objectives: To introduce conceptual knowledge on advanced statistical methods, chemical equilibria, and stoichiometry.	
1. Analytical chemometrics: Propagation of measurement uncertainties (inaccuracy and imprecision). Useful statistical test: test of significance, the F test, the student 't' test, the chi-test, the correlation coefficient, confidence limit of the mean, comparison of two standard values, comparison of standard deviation with average deviation, comparison of mean with true values, significant figures, regression analysis (least square method for linear and non-linear plots), statistics of sampling and detection limit evaluation. Chemometrics for modeling and parameter estimation, factor analysis, resolution and pattern recognition.	15
2. Treatment of Equilibria: Solvents and solutions, general treatment of equilibria in aqueous medium involving monoprotic weak acid and weak base, and salts of weak acids and weak bases. Activity and concentration, Effect of electrolytes on chemical equilibria, Calculation of pH, Constructing titration curves from charge balance and mass balance equations, Acid-base titrations and theory of pH indicators, Complexation equilibria and complexometric titrations, Redox equilibria and redox titration, Theory of redox indicators, Precipitation reaction and precipitation titrations and theory of adsorption indicators.	8
3. Spectrophotometric Determination of Stoichiometry of Complexes: 7 Job's method of continuous variation, mole ratio and slope ratio analysis, Advantages and limitations, typical examples.	7

Outcomes: The students will establish knowledge on chemometrics, theory of various titrations, and stoichiometric calculations using instrumental techniques.	
Suggested Books: 1. R. L. Pecsok, L. D. Shields, T. Cairns and L.C. Mc William, Modern Methods of Chemical Analysis, 2nd Edition (1976), John Wiley, New York. 2. G. D. Christian, Analytical Chemistry, 5th Edition (1994), John Wiley & Sons, New York. 3. D. A. Skoog, D.M. West, F.J. Holler, S.R. Crouch, Analytical Chemistry - An Introduction, 7th Edition (2000), Saunders College Publishing, Philadelphia, London. 4. J. H. Kennedy, Analytical Chemistry: Principles, 2nd Edition (1990), Saunders Holt, London.	

CHEMJ72: Inorganic Chemistry-V	
Objectives: 1. To train students about the illustration of symmetry in complexes and the fundamentals of character tables; and Orgel diagrams of metal complexes. 2. To teach the biological perspectives of important metal ions.	
1. Symmetry-based concepts of energy level diagrams of metal complexes: Symmetry elements and operations, determination of point group of a molecule, Group representations, Features of specific character tables in reference to Orgel diagrams, Symmetry-based concepts of Orgel energy level diagrams of metal complexes. Introduction to Tanabe-Sugano diagrams (d ² configuration)	15
2. Bioinorganic chemistry: Metal ions in biological processes. Biological role of iron. Oxygen transport with reference to hemoglobin and myoglobin. Synthetic model complexes for oxygen transport. Cytochromes and iron-sulfur proteins: Electron Transfer reactions. Photosystems, Role of metalloenzymes. Biological role of alkali metal ions, and their transport with special reference to Na ⁺ /K ⁺ ATPase. Vitamin B12. Nitrogen Fixation.	15
Outcomes: Students shall be able to 1. Learn the importance of symmetry-based concepts in the field of coordination chemistry. 2. Develop an interdisciplinary approach to metal chemistry with the biological world.	
Suggested Books: 1. Physical Methods in Inorganic Chemistry, R. S. Drago, International Edn. (1971), Affiliated East-West Press, New Delhi. 2. Chemical Applications of Group Theory, F. A. Cotton 3. Inorganic Chemistry, J.E. Huheey, E.A. Keiter and R.L. Keiter, 4th Edn. (1993), Addison-Wesley Pub. Co., New York. 4. Bioinorganic Chemistry, I. Bertini, H. B. Gray, S. J. Lippard, J.S. Valentine, Viva Books, P. Ltd. New Delhi, South Asian Ed., 1998 (Reprint 2004). 5. Principles of Bioinorganic Chemistry, S. J. Lippard, J. M. Berg, University Science Books, California 1994. 6. Bioinorganic Chemistry: Inorganic Elements in the Chemistry of Life, W. Kaim, B. Schwederski, John Wiley & Sons, England, 2004.	

CHEMJ73: Organic Chemistry-V	
Objectives: To impart knowledge of mechanistic aspects of the reactivity of C=X (X = C, N, O, S) multiple bonds, and aromatic nucleophilic substitution reactions.	
1. Nucleophilic Substitution and addition reactions: Mechanism and Stereochemistry of SN1, SN2, SNi & SN2' reactions. The effect of substrate structure, nucleophile, leaving group, temperature, and reaction medium. Phase transfer catalysis (Basic concept); epoxidation of unsaturated ketone, nucleophilic addition to C=X bond, solid-liquid phase transfer catalysis.	10
2. Elimination reactions: The E1, E2 and E1cB mechanisms, Orientation of the double bond. Hofmann versus Saytzeff elimination, Pyrolytic syn-elimination, Competition between substitution and elimination reactions. Stereoselective olefination reactions: Peterson, Julia and Petasis. Olefin metathesis (RCM) using Ru catalysis.	10
3. Neighbouring Group Participation: Evidences of N.G.P.; the phenonium ion, participation by π and σ bonds. Classical versus non-classical carbonium ions.	4
4. Aromatic nucleophilic substitution (SNAr): Benzyne Mechanism, Reactivity and effect of leaving group. Chichibabin reaction, von Richter rearrangement, Sommelet-Hauser rearrangement, and Smile rearrangement	6
Outcomes: The students will be able to understand 1. The mechanistic details of substitution and elimination reactions. 2. Reaction conditions for the formation of different ion intermediates. 3. Mechanistic aspects of some reaction intermediates. 4. Advantages and disadvantages of protection and deprotection approaches in organic syntheses.	
Suggested Books: 1. Clayden, Greeves, Warren and Wothers, Organic Chemistry, Oxford University Press, 2001. 2. Organic Chemistry", 9th Edition, L. G. Wade, J. W. Simek & M. S. Singh, Pearson India Education Services Pvt. Ltd. 2020 (ISBN 978-93-893-4267-3). 3. M. B. Smith & Jerry March, March's Advanced Organic Chemistry, 5th Edition (2001), John Wiley & Sons, New York. 4. Peter Sykes, A Guide book to Mechanism in Organic Chemistry, 6th Edition (1997), Orient Longman Ltd., New Delhi. 5. S. M. Mukherjee and S. P. Singh, Reaction Mechanism in Organic Chemistry, 1st Edition (1990), Macmillan India Ltd., New Delhi. 6. G. S. Zweifel and M. H. Nantz, Modern Organic Synthesis, (2007), Freeman and Company, New York.	

CHEM74: Physical Chemistry-V	
Objectives: To impart basic knowledge on energy levels in atom and molecules, atomic term symbols, and chemical bonding.	
1. Atomic structure: The Hydrogen atom, atomic orbitals, variation theorem, Secular determinant, He-atom, electron spin, Slater determinant, antisymmetry rule, and Pauli's exclusion principle, many-electron atoms: SCF method, atomic term symbols	15
2. Molecular structure: Born-Oppenheimer approximation, molecular orbital treatment for H_2^+ MO treatment of homo- and heteronuclear diatomic molecules, electronic terms of diatomic molecules, VB approach, comparison of the MO and VB theories, Hückel MO treatment of simple and conjugated polyenes.	15

Outcomes: After the completion of the course, the students should be able to 1. Gain a preliminary notion about the quantum mechanical model of the atom and the shape of s, p, d, and f-orbitals. 2. Explain the idea of spin quantum number, Slater determinant, antisymmetry rule and Pauli's exclusion principle and atomic term symbols term symbol. 3. Explain different energy level in molecule and rationalize the qualitative idea of bonding between atoms that generates molecules.	
Suggested Books: 1. Molecular Quantum Mechanics, P. W. Atkins and R. S. Friedman, 3rd edition (1997), Oxford University Press. Oxford. 2. Quantum Chemistry, I. N. Levine, 5th edition (2000), Pearson Educ., Inc., New Delhi. 3. Introductory Quantum Chemistry, A. K. Chandra, 4th Edn. (2017), McGraw Hill Education (India) Private Limited. 4. Quantum Chemistry, D. A. McQuarrie, Viva Books Private Limited, New Delhi, 2007.	

CHEMJ75: Material Chemistry	
Objectives: To impart knowledge on new materials and their classifications.	
1. Introduction: Materials and their classification, inorganic and organic materials.	4
2. Inorganic materials: Design and synthesis of inorganic materials, requirements and constraints, combination properties of composites, solid state reactions for synthesis of inorganic materials: ceramic methods, precursor method and sol-gel synthesis, physical and chemical vapour depositions; carbides, nitrides, structural and functional ceramics, intermetallics; ceramic superconductors, magnetic ceramic	10
3. Organic materials: Molecular electronics: molecular materials for electronics and molecular scale electronics: Molecular properties, molecular arrangement. and molecular interactions, piezoelectric and pyroelectric organic materials; molecular magnets based on transition metal complexes and organic ferromagnets, organic nonlinear optical materials: photochromic organic materials and their classes; conducting polymers: polyacetylene, polypyrrole, polyaniline, and polythiophene; conductive charge transfer materials: TTFTCNQ, metal-dithiolate systems, fullerenes. Langmuir Blodgett films, molecular electronic logic, and architectures.	10
4. Nanomaterials: Fundamentals of nanoscience: definition, nano versus bulk, quantum confinement: nanoscale in 1D, 2D, and 3D with examples, synthesis of nanomaterials: top-down and bottom-up approaches, size and shape-dependent optical properties of semiconducting (CdSe) and plasmonic metal nanoparticles, nanoclusters, and nanowires, nanoparticles, concept of magnetic nanoparticles, applications of nanomaterials.	6
Outcomes: The students will be able to clarify the concepts of molecular electronics.	
Suggested Books: 1. Inorganic Chemistry of Materials, P.J. Vander Put, Plenum Press, New York, 1998. 2. An Introduction to Molecular Electronics, M. C. Petty, M. R. Bryce and D. Bloor, Editors Edward Arnold, London 1995. 3. An Introduction to Nanomaterials and Nanoscience, A. K. Das, Mahua Das, CBS publishers and distributors Pvt Ltd (2019). 4. Characterization Techniques for Nanomaterials, I. Munaweera, M. L. Chamalki Madhusa, 7th Ed., CRC Press (2023).	

CHEMJ76: Physical Chemistry Lab Work	
1. Study of the kinetics of inversion of cane sugar catalysed by acid polarimetrically. 2. Verification of Debye-Hückel-Onsager equation $\Lambda = \Lambda_0 - S C^{1/2}$ for strong uni-univalent electrolyte. 3. Energy of activation of acid catalyzed hydrolysis of methyl acetate. 4. To study the effect of ionic strength on the reaction between $S_2O_8^{2-}$ and I^- ions. 5. Conductometric titration of a triple mixture ($HCl + NH_4Cl + KCl$) with (i) $NaOH$ and (ii) $AgNO_3$. 6. Verification of the Nernst law of electrode potential. 7. Potentiometric titration of a polybasic acid.	

Note: Experiments may be added/ modified subject to availability of facilities.	
CHEMJ77: Inorganic Chemistry Lab Work	
1. Quantitative estimation of metal ions (two) by gravimetric/ titrimetric/ spectrophotometric methods	
2. Study of symmetry elements and point groups using models	
Note: Experiments may be added/ modified subject to availability of facilities.	
Outcomes: Students shall be able to	
1. Learn the difference between thermodynamic and kinetic approach in the formation of inorganic complexes.	
2. Learn the ability for interpretation of IR, Raman, UV-vis, NMR and Mass spectral data.	
Suggested Books:	
1. F. Basalo and R. G. Pearson, Mechanism of Inorganic Reactions, 2nd Edn (1967), Wiley Eastern Ltd., New Delhi.	
2. D. F. Shriver and P. W. Atkins, Inorganic Chemistry, 3rd Edn. (1999), ELBS, London.	
3. F. A. Cotton and G. Wilkinson, Advanced Inorganic Chemistry, 6th Edn. (1999), John Wiley & Sons, New York.	
4. D.N. Sathyanarayana, Electronic Absorption Spectroscopy and Related Techniques, Universities Press (India) Ltd., Hyderabad (2001).	
5. Keith F. Purcell and John C. Kotz, Inorganic Chemistry, W. B. Saunders Com. (1987), Hong Kong.	
6. Martin L. Tobe and John Burgess, Inorganic Reaction Mechanisms, Longmans 1st Edn. (1999).	
7. D. Banerjea, M.K.Bharty, Mechanisms of Reactions of Metal Complexes in Solution, RSC, 2022.	

CHEMJ78: Organic Chemistry Lab Work	
Separation of Organic mixture and their identification	
Note: Experiments may be added/ modified subject to availability of facilities.	

CHEMJ81: Analytical Chemistry-IV	
Objectives: To impart knowledge on electrochemical methods, separation methods and thermal analysis.	
1. Electrochemical methods: Origin of polarography, Current-voltage relationship, Theory of polarographic waves (DC and sampled DC (tast) polarograms), Instrumentation, Ilkovic equation, Qualitative and quantitative applications.	8
2. Separation Methods: Principle of chromatography, Classifications of chromatography, Techniques of planar and column chromatography, Gas chromatography, High-performance liquid chromatography .	15
3. Thermal Analysis: Theory, methodology and applications of thermogravimetric analysis (TGA), Differential Thermal Analysis (DTA), and Differential scanning calorimetry (DSC). Principles, techniques and applications of thermometric titration methods.	7
Outcomes: The student will have deep understanding on the various analytical methods like polarography, column chromatography, TGA, DTA, and DSC.	
Suggested Books:	
1. D.A. Skoog, <i>Principles of Instrumental Analysis</i> , 5th Edition (1998), Saunders College Publishing, Philadelphia, London.	
2. G.W. Ewing, <i>Instrumental Methods of Chemical Analysis</i> , 5th Edition (1978), McGraw Hill Books Co., New York.	
3. R.L. Pecsok, L. D. Shields, T. Cairns and L.C. Mc William, <i>Modern Methods of Chemical Analysis</i> , 2nd Edition (1976), John Wiley, New York.	
4. J.H. Kennedy, <i>Analytical Chemistry: Principles</i> , 2nd Edition (1990), Saunders Holt, London.	

CHEMJ82: Inorganic Chemistry-VI	
Objectives: 1. To gain the concept of thermodynamic and kinetic stability of coordination complexes and inorganic substitution reactions to obtain complexes with selective stereo specificity. 2. To train the students about the characterization of inorganic compounds with the help of IR, NMR, UV-vis and Mass spectral techniques.	
1. Kinetics and Mechanism of Substitution Reactions: Nature of substitution reactions; prediction of reactivity of octahedral, tetrahedral and square-planar complexes in terms of crystal field activation energy and structure preference energy; rates of reactions; acid hydrolysis, base hydrolysis and anation reactions.	10
2. Electron Transfer Reactions: Mechanism and rate laws; various types of electron transfer reactions, Marcus-Hush theory, correlation between thermal and optical electron transfer reactions; identification of intervalence transfer bands in solution.	10
3. Characterization of Inorganic Compounds: Vibrational, NMR, Uv-vis, and Mass spectroscopic techniques	10
Outcomes: Students shall be able to 1. Learn the difference between thermodynamic and kinetic approach in the formation of inorganic complexes. 2. Learn the ability for interpretation of IR, Raman, UV-vis, NMR and Mass spectral data.	
Suggested Books: 1. F. Basalo and R. G. Pearson, <i>Mechanism of Inorganic Reactions</i> , 2 nd Edn (1967), Wiley Eastern Ltd., New Delhi. 2. D. F. Shriver and P. W. Atkins, <i>Inorganic Chemistry</i> , 3 rd Edn. (1999), ELBS, London. 3. F. A. Cotton and G. Wilkinson <i>Advanced Inorganic Chemistry</i> , 6 th Edn. (1999), John Wiley & Sons, New York. 4. D.N. Sathyanarayana, <i>Electronic Absorption Spectroscopy and Related Techniques</i> , Universities Press (India) Ltd., Hyderabad (2001). Keith F. Purcell and John C. Kotz, <i>Inorganic Chemistry</i> , W. B. Saunders Com.(1987), Hong Kong. 5. Martin L. Tobe and John Burgess, <i>Inorganic Reaction Mechanisms</i> , Longmans 1 st Edn. (1999). 6. D. Banerjee, M.K. Bharty, <i>Mechanisms of Reactions of Metal Complexes in Solution</i> , RSC, 2022	

CHEMJ83: Organic Chemistry-VI	
Objectives: The mechanistic aspects of asymmetric synthesis, pericyclic reactions, and the formation of carbon-carbon bonds in contemporary organic synthesis.	
1. Asymmetric Synthesis: Enolization (Lithium and Boron enolate), Stereoselective Aldol condensation reaction (chiral enolate & achiral aldehyde and achiral enolate and chiral aldehyde). The chiral pool, chiral auxiliaries, chiral catalysts and reagents, and enantiomeric excess.	8
2. Pericyclic Reactions: Woodward-Hoffmann rules and molecular orbitals; Electrocyclic reactions; Cycloaddition reactions, 1,3-Dipolar cycloaddition and cheletropic reactions; Sigmatropic rearrangements; Group transfer reactions: Ene reactions.	8
3. Modern organic synthesis: Retrosynthesis, Umpolung, Domino reactions and functional group interconversion.	6
4. Formation of Carbon-Carbon Bonds via Organometallic reagents: Organozinc, Organocopper, and Organoborane reagents, Palladium catalyzed coupling reactions (Heck, Sonogashira, and Suzuki).	8
Outcomes: The students will be able to understand 1. the mechanistic details of asymmetric synthesis of some bioactive molecules. 2. C-C Bond formation employing pericyclic reactions and organometallic reagents. 3. mechanistic aspects some functional group interconversion.	
Suggested Books: 1. Clayden, Greeves, Warren and Wothers, <i>Organic Chemistry</i> , Oxford University Press, 2001. 2. <i>Advanced Organic Chemistry-Reactions and Mechanisms 2e</i> —Maya Shankar Singh & Keshav Raghuvanshi, Pearson Education Pvt. Ltd., 2021 (ISBN 978-93-544- 9078-1)	

3.	M.B. Smith & Jerry March, March's Advanced Organic Chemistry, 5th Edition (2001), John Wiley & Sons, New York.	
4.	G. S. Zweifel and M. H. Nantz, Modern Organic Synthesis, (2007), Freeman and Company, New York.	
5.	S. Kumar, V. Kumar, and S. P. Singh, Pericyclic Reaction; A Mechanistic and Problem-Solving Approach, 1st Edition (2015), Elsevier Inc.	
6.	T. H. Lowry and K. S. Richardson, Mechanism and Theory in Organic Chemistry, 3rd Edition (1998), Addison – Wesley Longman Inc. (IS Edition).	
7.	I. Fleming, Pericyclic Reactions, Oxford University Press, Oxford (1999).	
8.	W. Carruthers and Iain Coldham Modern Methods of Organic Synthesis Cambridge University Press	
9.	Charles H. Depuy and Orville L. Chapman, Molecular reactions and photochemistry, New Delhi: Prentice Hall of India, 1988.	

CHEMJ84:Physical Chemistry-VI		
Objectives: To impart knowledge on different advanced electrochemistry topics (like electrochemical double layer, electrode kinetics and cyclic voltammetry), basic concepts of polymer chemistry and micelles and colloids.		
1. Advanced Electrochemistry: Metal/Electrolyte interface: OHP and IHP, potential profile across double layer region, potential difference across electrified interface; structure of the double layer: Helmholtz- Perrin, Gouy-Chapman, and Stern models. Butler-Volmer equation under near equilibrium and non-equilibrium conditions, exchange current density, Tafel plot, polarizable and non-polarizable interfaces. Semiconductor (SC)/electrolyte interface: Creation of space charge region, capacity of space-charge, Mott-Schottky plots for n-type and p-type semiconductors, determination of flat-band potential and donor/acceptor densities, application of SC/electrolyte interface in solar cells, Fuel Cell, Battery. Cyclic voltammetry (CV) technique basics and applications.		15
2. Polymer Chemistry: Introduction, Classification of Polymers & Intermolecular forces of Polymers. Different types of molecular weight of polymer, polydispersity Index, Determination of molecular weight of polymers. Thermodynamic Aspects of Polymerization. Mechanism of Living Radical Polymerizations: Nitroxide mediated polymerization (NMP), Metal-catalyzed Living Radical Polymerization, Reversible Addition-Fragmentation Chain Transfer (RAFT) Radical Polymerization. Coordination polymerization, Ring-opening polymerization. Thermal and mechanical properties of polymers: Glass transition temperature, melting temperature, Rheological properties, industrial polymers.		8
3. Micelles and Colloids: Surface active agents and their classification, micellization, hydrophobic interaction, critical micellar concentration (cmc), factors affecting cmc of surfactants, thermodynamics of micellization, phase separation and mass action models, reverse micelles, solubilization. Stability and properties of colloids. Zeta potential, Stern double layer, Tyndall effect.		7
Outcomes: After the completion of the course, the students should be able to Explain the origin of electrical double layer, different models describing double layer and electrical double layer capacitance and their applications. 1. Understand the kinetics of electrode reaction, exchange current density, overpotential, transference coefficient, Tafel plot. 2. Know definition of polymer, molecular weight, different type of polymerization techniques with emphasis of radical polymerization, their kinetics, thermal and mechanical properties of polymer. 3. Understand the fundamentals of micelles and colloids, CMC, stability and properties of colloids, Zeta potential, Stern double layer, Tyndall effect.		
Suggested Books: 1. Modern Electrochemistry, Vol. 2 A & B, J.O'M. Bockris and A. K. N. Reddy, Plenum Press, New York. 2. F. W. Billmeyer, Jr., Text Book of Polymer Science, 3rd Edition (1984), Wiley Interscience, New York. 3. Fundamentals of Polymerization, B. M. Mandal, Illustrated edition (2012), World Scientific; Singapore. 4. Micelles: Theoretical and Applied Aspects, Y. Moroi, Plenum Press, New York		

CHEMJ85: Advanced Spectroscopy	
Objectives: The course aims to provide the knowledge on basic fundamentals of various spectroscopic techniques like UV-Vis, IR, NMR, Raman, Fluorescence, Electron Paramagnetic Resonance spectroscopy, Mass spectrometry and Lasers.	
1. UV-Vis spectroscopy: π - π^* and n - π^* electronic transitions, effects of substituents, conjugation and chain length (on λ_{max} & ϵ_{max}) of conjugated polyenes and polyones, Woodward-Feiser rules for calculating λ_{max} .	3
2. IR spectroscopy: Molecular vibrations and IR absorption, the functional groups and fingerprint regions, the effect of bond orders on C-C, C-O, C-N bands; ring size and electronic effects. H-bonding and dilution effect.	4
3. NMR spectroscopy: The chemical shifts, diamagnetic shielding, anisotropic effect, ^1H - ^1H spin-spin splitting, and coupling constant, Interpretation of NMR spectra (^1H & ^{13}C NMR).	4
4. Mass spectrometry: Basics, mass spectra of straight and branched chain alkanes, molecular ion, base and isotope peaks, fragmentation pattern of alkyl halides, ethers, alcohols and ketones, McLafferty rearrangement.	2
5. Identification of simple organic molecules using spectral data (UV-Vis, IR, NMR, and Mass).	2
6. Raman spectroscopy: Rayleigh Scattering, Raman scattering effect, classical treatment of Raman scattering, rotational Raman transitions and rotation-vibrational Raman transitions.	3
7. Fluorescence spectroscopy: Instrumentation, fluorophores, solvent and environmental effect on fluorescence, quenching of fluorescence, fluorescence anisotropy, time-resolved fluorescence spectroscopy (TCSPC).	4
8. Lasers: laser action, characteristics, pulsed lasers, non-linear effects, harmonic generation, examples of lasers: He-Ne, Nd-YAG. Single molecule spectroscopy: Principle, instrumentation and application.	4
9. Electron Paramagnetic Resonance Spectroscopy: Introduction to electron spin resonance, Hyperfine splitting, Factors affecting hyperfine splitting, EPR of hydrogen, organic radicals in solution.	4
Outcomes: After completion of the course, the student should be able to	
1. Explain the principles of UV-Vis, IR, NMR, Raman, Electron Paramagnetic Resonance spectroscopy and interpret the spectral data for simple molecular system. 2. Understand the basic instrumentation of fluorimeter and some advance topics on fluorescence spectroscopy like solvent polarity effect on fluorescence, mode of interaction between fluorophores and quenchers, fluorescence anisotropy and fluorescence lifetime and its measurement. 3. Explore the theories Mass spectrometry to realize the spectra and characterize simple/complex organic molecules. 4. Understand basic principle of EPR and analyse the EPR spectra of systems having unpaired electrons. 5. Understand LASER and basics of single molecule spectroscopy and their application.	
Suggested Books:	
1. Principles of Fluorescence Spectroscopy, Joseph R. Lakowicz, 2006, Springer. 2. Modern Spectroscopy, J. M. Hollas, 4th Edn., (2004), John Wiley & Sons, Inc, New York. 3. Atkins's Physical Chemistry by Peter Atkins and Julio de Paula, W. H. Freeman & Co. 4. Fundamentals of Molecular Spectroscopy, C. N. Banwell, 4th Edn. (2017), Paperback, McGraw Hill Education (India) Private Limited, New Delhi. 5. Physical methods in Inorganic Chemistry by Russell S. Drago (Affiliated East- West Press Pvt. New Delhi). 6. Introduction to Magnetic Resonance by A. Carrington and A. D. McLachlan. (Chapman & Hall. London)	

CHEMJ86: Physical Chemistry Lab Work	
1. Determination of the rate constant for the decomposition of benzene diazonium chloride by gasometric method. 2. Degree of hydrolysis of urea-hydrochloride by kinetics method. 3. Conductometric study of the kinetics of saponification of ethyl acetate. 4. Determination of Critical Micelle Concentration (CMC) of a Surfactant by conductometry. 5. Electrochemical Impedance study of metal/solution interface. 6. Cyclic Voltammetry of the $[\text{Fe}(\text{CN})_6]^{3-}/[\text{Fe}(\text{CN})_6]^{4-}$ system. 7. Analysis of halide mixture by differential potentiometry. 8. Fluorescence quenching of organic fluorophores. 9. FRET for suitable donor – acceptor pair. 10. Determination of molecular weight by viscometry: Polyacrylamide-aq. NaNO_2 solution/Polyvinyl pyrrolidone in water. 11. Analysis of the given (i) vibrational-rotational, (ii) FTIR, and (iii) NMR spectra of a Compound	
Note: Experiments may be added/ modified subject to availability of facilities.	

CHEMJ87: Inorganic Chemistry Lab Work	
1. Synthesis of some coordination complexes (Including those involving more than one step). 2. Conductivity and spectroscopic studies of the complexes	
Note: Experiments may be added/ modified subject to availability of facilities.	

CHEMJ88: Organic Chemistry Lab Work	
Two-step synthesis of organic compounds: 1. Preparation of m-nitroaniline from nitrobenzene 2. Preparation of anthranilic acid from phthalic anhydride. 3. Preparation of 2-phenylindole from phenylhydrazine. 4. Preparation of methyl red dye 5. Preparation of thioxazole	
Note: Experiments may be added/ modified subject to availability of facilities.	

Semester- VII (Honours with Research)

CHEMJ7R1: Analytical Chemistry-III	
Objectives: To introduce conceptual knowledge on advanced statistical methods, chemical equilibria, and stoichiometry.	
1. Analytical chemometrics: Propagation of measurement uncertainties (inaccuracy and imprecision). Useful statistical test: test of significance, the F test, the student 't' test, the chi-test, the correlation coefficient, confidence limit of the mean, comparison of two standard values, comparison of standard deviation with average deviation, comparison of mean with true values, significant figures, regression analysis (least square method for linear and non-linear plots), statistics of sampling and detection limit evaluation. Chemometrics for modeling and parameter estimation, factor analysis, resolution and pattern recognition.	15
2. Treatment of Equilibria: Solvents and solutions, general treatment of equilibria in aqueous medium involving monoprotic weak acid and weak base, and salts of weak acids and weak bases. Activity and concentration, Effect of electrolytes on chemical equilibria, Calculation of pH, Constructing titration curves from charge balance and mass balance equations, Acid-base titrations and theory of pH indicators, Complexation equilibria and complexometric titrations, Redox equilibria and redox titration, Theory of redox indicators, Precipitation reaction and precipitation titrations and theory of adsorption indicators.	8
3. Spectrophotometric Determination of Stoichiometry of Complexes: Job's method of continuous variation, mole ratio and slope ratio analysis, Advantages and limitations, typical examples.	7
Outcomes: The students will establish knowledge on chemometrics, theory of various titrations, and stoichiometric calculations using instrumental techniques.	
Suggested Books: 1. R. L. Pecsok, L. D. Shields, T. Cairns and L.C. Mc William, Modern Methods of Chemical Analysis, 2nd Edition (1976), John Wiley, New York. 2. G. D. Christian, Analytical Chemistry, 5th Edition (1994), John Wiley & Sons, New York. 3. D. A. Skoog, D.M. West, F.J. Holler, S.R. Crouch, Analytical Chemistry - An Introduction, 7th Edition (2000), Saunders College Publishing, Philadelphia, London. 4. J. H. Kennedy, Analytical Chemistry: Principles, 2nd Edition (1990), Saunders Holt, London.	

CHEMJ7R2: Inorganic Chemistry-V	
Objectives: 1. To train students about the illustration of symmetry in complexes and the fundamentals of character tables; and Orgel diagrams of metal complexes. 2. To teach the biological perspectives of important metal ions.	
1. Symmetry-based concepts of energy level diagrams of metal complexes: Symmetry elements and operations, determination of point group of a molecule, Group representations, Features of specific character tables in reference to Orgel diagrams, Symmetry-based concepts of Orgel energy level diagrams of metal complexes. Introduction to Tanabe-Sugano diagrams (d ² configuration)	15
2. Bioinorganic chemistry: Metal ions in biological processes. Biological role of iron. Oxygen transport with reference to hemoglobin and myoglobin. Synthetic model complexes for oxygen transport. Cytochromes and iron-sulfur proteins: Electron Transfer reactions. Photosystems, Role of metalloenzymes. Biological role of alkali metal ions, and their transport with special reference to Na ⁺ /K ⁺ ATPase. Vitamin B12. Nitrogen Fixation.	15
Outcomes: Students shall be able to 1. Learn the importance of symmetry-based concepts in the field of coordination chemistry. 2. Develop an interdisciplinary approach to metal chemistry with the biological world.	
Suggested Books: 1. Physical Methods in Inorganic Chemistry, R. S. Drago, International Edn. (1971), Affiliated East-West Press, New Delhi. 2. Chemical Applications of Group Theory, F. A. Cotton 3. Inorganic Chemistry, J.E. Huheey, E.A. Keiter and R.L. Keiter, 4th Edn. (1993), Addison-	

4.	Wesley Pub. Co., New York.	
5.	Bioinorganic Chemistry, I. Bertini, H. B. Gray, S. J. Lippard, J.S. Valentine, Viva Books, P. Ltd. New Delhi, South Asian Ed., 1998 (Reprint 2004).	
6.	Principles of Bioinorganic Chemistry, S. J. Lippard, J. M. Berg, University Science Books, California 1994.	
	Bioinorganic Chemistry: Inorganic Elements in the Chemistry of Life, W. Kaim, B. Schwederski, John Wiley & Sons, England, 2004.	

CHEMJ7R3: Organic Chemistry-V		
Objectives: To impart knowledge of mechanistic aspects of the reactivity of C=X (X = C, N, O, S) multiple bonds and aromatic nucleophilic substitution reactions.		
1. Nucleophilic Substitution and addition reactions: Mechanism and Stereochemistry of SN1, SN2, SNi & SN2' reactions. The effect of substrate structure, nucleophile, leaving group, temperature, and reaction medium. Phase transfer catalysis (Basic concept); epoxidation of unsaturated ketone, nucleophilic addition to C=X bond, solid-liquid phase transfer catalysis.	10	
2. Elimination reactions: The E1, E2 and E1cB mechanisms, Orientation of the double bond. Hofmann versus Saytzeff elimination, Pyrolytic syn-elimination, Competition between substitution and elimination reactions. Stereoselective olefination reactions: Peterson, Julia and Petasis. Olefin metathesis (RCM) using Ru catalysis.	10	
3. Neighbouring Group Participation: Evidences of N.G.P.; the phenonium ion, participation by π and σ bonds. Classical versus non-classical carbonium ions.	4	
4. Aromatic nucleophilic substitution (SNAr): Benzyne Mechanism, Reactivity and effect of leaving group. Chichibabin reaction, von Richter rearrangement, Sommelet-Hauser rearrangement, and Smile rearrangement	6	
Outcomes: The students will be able to understand 1. The mechanistic details of substitution and elimination reactions. 2. Reaction conditions for the formation of different ion intermediates. 3. Mechanistic aspects of some reaction intermediates. 4. Advantages and disadvantages of protection and deprotection approaches in organic syntheses.		
Suggested Books: 1. Clayden, Greeves, Warren and Wothers, Organic Chemistry, Oxford University Press, 2001. 2. Organic Chemistry", 9th Edition, L. G. Wade, J. W. Simek & M. S. Singh, Pearson India Education Services Pvt. Ltd. 2020 (ISBN 978-93-893-4267-3). 3. M. B. Smith & Jerry March, March's Advanced Organic Chemistry, 5th Edition (2001), John Wiley & Sons, New York. 4. Peter Sykes, A Guide book to Mechanism in Organic Chemistry, 6th Edition (1997), Orient Longman Ltd., New Delhi. 5. S. M. Mukherjee and S. P. Singh, Reaction Mechanism in Organic Chemistry, 1st Edition (1990), Macmillan India Ltd., New Delhi. 6. G. S. Zweifel and M. H. Nantz, Modern Organic Synthesis, (2007), Freeman and Company, New York.		

CHEMJ7R4: Physical Chemistry-V		
Objectives: To impart basic knowledge on energy levels in atom and molecules, atomic term symbols, and chemical bonding.		
1. Atomic structure: The Hydrogen atom, atomic orbitals, variation theorem, Secular determinant, He-atom, electron spin, Slater determinant, anti symmetry rule, and Pauli's exclusion principle, many-electron atoms: SCF method, atomic term symbols	15	

2. Molecular structure: Born-Oppenheimer approximation, molecular orbital treatment for H_2^+ MO treatment of homo- and heteronuclear diatomic molecules, electronic terms of diatomic molecules, VB approach, comparison of the MO and VB theories, Hückel MO treatment of simple and conjugated polyenes .	15
Outcomes: After the completion of the course, the students should be able to 1. Gain a preliminary notion about the quantum mechanical model of the atom and the shape of s, p, d, and f-orbitals. 2. Explain the idea of spin quantum number, Slater determinant, anti symmetry rule and Pauli's exclusion principle and atomic term symbols term symbol. 3. Explain different energy level in molecule and rationalize the qualitative idea of bonding between atoms that generates molecules.	
Suggested Books: 1. Molecular Quantum Mechanics, P. W. Atkins and R. S. Friedman, 3rd edition (1997), Oxford University Press. Oxford. 2. Quantum Chemistry, I. N. Levine, 5th edition (2000), Pearson Educ., Inc., New Delhi. 3. Introductory Quantum Chemistry, A. K. Chandra, 4thEdn. (2017), McGraw Hill Education (India) Private Limited. 4. Quantum Chemistry, D. A. Mc Quarrie, Viva Books Private Limited, New Delhi, 2007.	
CHEMJ7R5: Material Chemistry	
Objectives: To impart knowledge on new materials and their classifications.	
1. Introduction: Materials and their classification, inorganic and organic materials.	4
2. Inorganic materials: Design and synthesis of inorganic materials, requirements and constraints, combination properties of composites, solid state reactions for synthesis of inorganic materials: ceramic methods, precursor method and sol-gel synthesis, physical and chemical vapour depositions; carbides, nitrides, structural and functional ceramics, intermetallics; ceramic superconductors, magnetic ceramics. .	10
3. Organic materials: Molecular electronics: molecular materials for electronics and molecular scale electronics: Molecular properties, molecular arrangement. and molecular interactions, piezoelectric and pyroelectric organic materials; molecular magnets based on transition metal complexes and organic ferromagnets, organic nonlinear optical materials: photochromic organic materials and their classes; conducting polymers: polyacetylene, polypyrrole, polyaniline, and polythiophene; conductive change transfer materials: TTFTCNQ, metal-dithiolate systems, fullerenes. LangmuirBlodgett films, molecular electronic logic, and architectures.	10
4. Nanomaterials: Fundamentals of nanoscience: definition, nano versus bulk, quantum confinement: nanoscale in 1D, 2D, and 3D with examples, synthesis of nanomaterials: top-down and bottom-up approaches, size and shape-dependent optical properties of semiconducting (CdSe) and plasmonic metal nanoparticles, nanoclusters, and nanowires, nanoparticles, concept of magnetic nanoparticles, applications of nanomaterials.	6
Outcomes: The students will be able to clarify the concepts of molecular electronics.	
Suggested Books: 1. Inorganic Chemistry of Materials, P.J. Vander Put, Plenum Press, New York, 1998. 2. An Introduction to Molecular Electronics, M. C. Petty, M. R. Bryce and D. Bloor, Editors Edward Arnold, London 1995. 3. An Introduction to Nanomaterials and Nanoscience, A. K. Das, Mahua Das, CBS publishers and distributors Pvt Ltd (2019). 4. Characterization Techniques for Nanomaterials, I. Munaweera, M. L. Chamalki Madhusa, 7th Ed., CRC Press (2023).	

CHEMJ7R6: Physical Chemistry Lab Work		
1.	Study of the kinetics of inversion of cane sugar catalysed by acid polarimetrically.	
2.	Verification of Debye-Hückel-Onsager equation $\Lambda = \Lambda_0 - S C^{1/2}$ for strong uni-univalent electrolyte.	
3.	Energy of activation of acid catalyzed hydrolysis of methyl acetate.	
4.	To study the effect of ionic strength on the reaction between $S_2O_8^{2-}$ and I^- ions.	
5.	Conductometric titration of a triple mixture (HCl+NH ₄ Cl+KCl) with (i) NaOH and (ii) AgNO ₃ .	
6.	Verification of the Nernst law of electrode potential.	
7.	Potentiometric titration of a polybasic acid.	
Note: Experiments may be added/ modified subject to availability of facilities.		

CHEMJ7R7: Inorganic Chemistry Lab Work		
1.	Quantitative estimation of metal ions (two) by gravimetric/ titrimetric/spectrophotometric methods	
2.	Study of symmetry elements and point groups using models	
Note: Experiments may be added/ modified subject to availability of facilities.		

CHEMJ7R8: Organic Chemistry Lab Work		
Separation of Organic mixture and their identification		
Note: Experiments may be added/ modified subject to availability of facilities.		

Semester-VIII (Total Credits 16; L-4, P-12)

The students will opt any one paper out of the four papers

CHEMJ8R1: Analytical

Chemistry-IV

CHEMJ8R2: Inorganic

Chemistry-VI

CHEMJ8R3: Organic

Chemistry-VI

CHEMJ8R4: Physical

Chemistry-VI

The course CHEMJ8R5: Advanced Spectroscopy is compulsory

CHEMJ8R1: Analytical Chemistry-IV	
Objectives: To impart knowledge on electrochemical methods, separation methods and thermal analysis.	
1. Electrochemical methods: Origin of polarography, Current-voltage relationship, Theory of polarographic waves (DC and sampled DC (tast) polarograms), Instrumentation, Ilkovic equation, Qualitative and quantitative applications.	8
2. Separation Methods: Principle of chromatography, Classifications of chromatography, Techniques of planar and column chromatography, Gas chromatography, High-performance liquid chromatography	15
3. Thermal Analysis: Theory, methodology and applications of thermogravimetric analysis (TGA), Differential Thermal Analysis (DTA), and Differential scanning calorimetry (DSC). Principles, techniques and applications of thermometric titration methods.	7
Outcomes: The student will have deep understanding on the various analytical methods like polarography, column chromatography, TGA, DTA, and DSC.	
Suggested Books: 1. D.A. Skoog, <i>Principles of Instrumental Analysis</i> , 5th Edition (1998), Saunders College Publishing, Philadelphia, London. 2. G.W. Ewing, <i>Instrumental Methods of Chemical Analysis</i> , 5th Edition (1978), McGraw Hill Books Co., New York. 3. R.L. Pecsok, L. D. Shields, T. Cairns and L.C. Mc William, <i>Modern Methods of Chemical Analysis</i> , 2nd Edition (1976), John Wiley, New York. 4. J.H. Kennedy, <i>Analytical Chemistry: Principles</i> , 2nd Edition (1990), Saunders Holt, London.	

CHEMJ8R2: Inorganic Chemistry-VI	
Objectives: 1. To gain the concept of thermodynamic and kinetic stability of coordination complexes and inorganic substitution reactions to obtain complexes with selective stereospecificity. 2. To train the students about the characterization of inorganic compounds with the help of IR, NMR, UV-vis and Mass spectral techniques.	
1. Kinetics and Mechanism of Substitution Reactions: Nature of substitution reactions; prediction of reactivity of octahedral, tetrahedral and square-planar complexes in terms of crystal field activation energy and structure preference energy; rates of reactions; acid hydrolysis, base hydrolysis and anation reactions.	10
2. Electron Transfer Reactions: Mechanism and rate laws; various types of electron transfer reactions, Marcus-Hush theory, correlation between thermal and optical electron transfer reactions; identification of inter valence transfer bands in solution.	10
3. Characterization of Inorganic Compounds: Vibrational, NMR, Uv-vis, and Mass spectroscopic techniques	10
Outcomes: Students shall be able to 1. Learn the difference between thermodynamic and kinetic approach in the formation of inorganic complexes. 2. Learn the ability for interpretation of IR, Raman, UV-vis, NMR and Mass spectral data.	
Suggested Books: 1. F. Basalo and R. G. Pearson, <i>Mechanism of Inorganic Reactions</i> , 2nd Edn (1967), Wiley Eastern Ltd., New Delhi. 2. D. F. Shriver and P. W. Atkins, <i>Inorganic Chemistry</i> , 3rd Edn. (1999), ELBS, London. 3. F. A. Cotton and G. Wilkinson, <i>Advanced Inorganic Chemistry</i> , 6th Edn. (1999), John Wiley & Sons, New York. 4. D.N. Sathyanarayana, <i>Electronic Absorption Spectroscopy and Related Techniques</i> , Universities Press (India) Ltd., Hyderabad (2001). 5. Keith F. Purcell and John C. Kotz, <i>Inorganic Chemistry</i> , W. B. Saunders Com. (1987), Hong Kong. 6. Martin L. Tobe and John Burgess, <i>Inorganic Reaction Mechanisms</i> , Longmans 1st Edn. (1999). 7. J. Singh, M.D. Pandey and J. Singh, Spectroscopy of Inorganic Compounds , New Age International, 2nd Ed, 2024.	

CHEMJ8R3: Organic Chemistry-VI	
Objectives: The mechanistic aspects of asymmetric synthesis, pericyclic reactions, and the formation of carbon-carbon bonds in contemporary organic synthesis.	
1. Asymmetric Synthesis: Enolization (Lithium and Boron enolate), Stereo selective Aldol condensation reaction (chiral enolate & achiral aldehyde and achiral enolate and chiral aldehyde). The chiral pool, chiral auxiliaries, chiral catalysts and reagents, and enantiomeric excess.	8
2. Pericyclic Reactions: Woodward-Hoffmann rules and molecular orbitals; Electrocyclic reactions; Cycloaddition reactions, 1,3-Dipolar cycloaddition and cheletropic reactions; Sigmatropic rearrangements; Group transfer reactions: Ene reactions.	8
3. Modern organic synthesis: Retrosynthesis, Umpolung, Domino reactions and functional group interconversion.	6
4. Formation of Carbon-Carbon Bonds via Organometallic reagents: Organozinc, Organocopper, and Organoborane reagents, Palladium catalyzed coupling reactions (Heck, Sonogashira, and Suzuki).	8
Outcomes: The students will be able to understand 1. The mechanistic details of asymmetric synthesis of some bioactive molecules. 2. C-C Bond formation employing pericyclic reactions and organometallic reagents. 3. Mechanistic aspects some functional group interconversion.	
Suggested Books: 1. Clayden, Greeves, Warren and Wothers, Organic Chemistry, Oxford University Press, 2001. 2. Advanced Organic Chemistry-Reactions and Mechanisms 2e—Maya Shankar Singh & Keshav Raghuvanshi, Pearson Education Pvt. Ltd., 2021 (ISBN 978-93-544- 9078-1) 3. M.B. Smith & Jerry March, March's Advanced Organic Chemistry, 5th Edition (2001), John Wiley & Sons, New York. 4. G. S. Zweifel and M. H. Nantz, Modern Organic Synthesis, (2007), Freeman and Company, New York. 5. S. Kumar, V. Kumar, and S. P. Singh, Pericyclic Reaction; A Mechanistic and Problem-Solving Approach, 1st Edition (2015), Elsevier Inc. 6. T. H. Lowry and K. S. Richardson, Mechanism and Theory in Organic Chemistry, 3rd Edition (1998), Addison – Wesley Longman Inc. (IS Edition). 7. I. Fleming, Pericyclic Reactions, Oxford University Press, Oxford (1999). 8. W. Carruthers and Iain Coldham Modern Methods of Organic Synthesis Cambridge University Press 9. Charles H. Depuy and Orville L. Chapman, Molecular reactions and photochemistry, New Delhi: Prentice Hall of India, 1988.	

CHEMJ8R4: Physical Chemistry-VI	
Objectives:	
1. Advanced Electrochemistry: Metal/Electrolyte interface: OHP and IHP, potential profile across double layer region, potential difference across electrified interface; structure of the double layer: Helmholtz- Perrin, Gouy-Chapman, and Stern models. Butler-Volmer equation under near equilibrium and non-equilibrium conditions, exchange current density, Tafel plot, polarizable and non-polarizable interfaces. Semiconductor (SC)/electrolyte interface: Creation of space charge region, capacity of space-charge, Mott-Schottky plots for n-type and p-type semiconductors, determination of flat-band potential and donor/acceptor densities, application of SC/electrolyte interface in solar cells, Fuel Cell, Battery. Cyclic voltammetry (CV) technique basics and applications.	15
2. Polymer Chemistry: Introduction, Classification of Polymers & Intermolecular forces of Polymers. Different types of molecular weight of polymer, polydispersity Index, Determination of molecular weight of polymers. Thermodynamic Aspects of Polymerization. Mechanism of Living Radical Polymerizations: Nitroxide mediated polymerization (NMP), Metal-catalyzed Living Radical Polymerization, Reversible Addition-Fragmentation Chain Transfer (RAFT) Radical Polymerization. Coordination polymerization, Ring-opening polymerization. Thermal and mechanical properties of polymers: Glass transition temperature, melting temperature, Rheological properties, industrial polymers.	8

3. Micelles and Colloids: Surface active agents and their classification, micellization, hydrophobic interaction, critical micellar concentration (cmc), factors affecting cmc of surfactants, thermodynamics of micellization, phase separation and mass action models, reverse micelles, solubilization. Stability and properties of colloids. Zeta potential, Stern double layer, Tyndal effect.	7
Outcomes:	
Suggested Books:	
1. Modern Electrochemistry, Vol. 2 A & B, J.O'M. Bockris and A. K. N. Reddy, Plenum Press, New York.	
2. F. W. Billmeyer, Jr., Text Book of Polymer Science, 3rd Edition (1984), Wiley Interscience, New York.	
3. Fundamentals of Polymerization, B. M. Mandal, Illustrated edition (2012), World Scientific; Singapore.	
4. Micelles: Theoretical and Applied Aspects, Y. Moroi, Plenum Press, New York	

CHEMJ8R5: Advanced Spectroscopy	
Objectives:	
1. UV-Vis spectroscopy: $\pi-\pi^*$ and $n-\pi^*$ electronic transitions, effects of substituents, conjugation and chain length (on $\lambda_{\max}$ & $\epsilon_{\max}$) of conjugated polyenes and polyones, Woodward-Feiser rules for calculating $\lambda_{\max}$.	3
2. IR spectroscopy: Molecular vibrations and IR absorption, the functional groups and fingerprint regions, the effect of bond orders on C-C, C-O, C-N bands; ring size and electronic effects. H-bonding and dilution effect.	4
3. NMR spectroscopy: The chemical shifts, diamagnetic shielding, anisotropic effect, $1H-1H$ spin-spin splitting, and coupling constant, Interpretation of NMR spectra (1H & ^{13}C NMR).	4
4. Mass spectrometry: Basics, mass spectra of straight and branched chain alkanes, molecular ion, base and isotope peaks, fragmentation pattern of alkyl halides, ethers, alcohols and ketones, McLafferty rearrangement.	2
5. Identification of simple organic molecules using spectral data (UV-Vis, IR, NMR, and Mass).	2
6. Raman spectroscopy: Rayleigh Scattering, Raman scattering effect, classical treatment of Raman scattering, rotational Raman transitions and rotation-vibrational Raman transitions.	3
7. Fluorescence spectroscopy: Instrumentation, fluorophores, solvent and environmental effect on fluorescence, quenching of fluorescence, fluorescence anisotropy, time-resolved fluorescence spectroscopy (TCSPC).	4
8. Lasers: laser action, characteristics, pulsed lasers, non-linear effects, harmonic generation, examples of lasers: He-Ne, Nd-YAG. Single molecule spectroscopy: Principle, instrumentation and application.	4
9. Electron Paramagnetic Resonance Spectroscopy: Introduction to electron spin resonance, Hyperfine splitting, Factors affecting hyperfine splitting, EPR of hydrogen, organic radicals in solution.	4
Outcomes:	
Suggested Books:	
1. Principles of Fluorescence Spectroscopy, Joseph R. Lakowicz, 2006, Springer.	
2. Modern Spectroscopy, J. M. Hollas, 4th Edn., (2004), John Wiley & Sons, Inc, New York.	
3. Atkins's Physical Chemistry by Peter Atkins and Julio de Paula, W. H. Freeman & Co.	
4. Fundamentals of Molecular Spectroscopy, C. N. Banwell, 4th Edn. (2017), Paperback, McGraw Hill Education (India) Private Limited, New Delhi.	
5. Physical methods in Inorganic Chemistry by Russell S. Drago (Affiliated East-West Press Pvt. New Delhi).	
6. Introduction to Magnetic Resonance by A. Carrington and A. D. McLachlan. (Chapman & Hall. London)	

CHEMN11: Basic Concepts of Chemistry-I	
Objectives: To understand the wave nature of matter and its implications. To study the classification of elements and their properties. To study the states of matter; behavior of ideal and real gases, liquids, and the structures of solids. To understand the laws of thermodynamics.	
1. Atomic Structure: Wave-particle duality, Heisenberg uncertainty principle, Schrodinger wave equation, Eigenvalues and Eigenfunctions, significance of Ψ and Ψ^2 Radial and angular wave functions, Multi-electron system: Concept of Effective Nuclear Charge (Slater's rules).	5
2. Atomic properties and their periodic trends: A brief introduction to the modern periodic table (Block wise). Atomic size (Covalent, Ionic and van der Waal's radii; Lanthanide contraction), Electron loss, and electron gain enthalpies. Inert pair effect. Electronegativity (Pauling, Mulliken, Allred Rochow's methods of determination): As a criterion of bond type. Lattice energy and its determination: a) Born-Haber cycle b) Born Lande equation c) Comparison of experimental and theoretical values. Solvation/Hydration energy (Solubility of ionic solids in water). Covalency in ionic compounds (Fajan's rules). Redox reaction, Standard electrode potential and its determination. Electrochemical series and its applications Latimer and Frost diagrams; specific illustrative examples.	10
3. Gaseous State: Kinetic theory of gases and ideal gas laws, Maxwell distribution of speed, Equipartition of Energy collisions in a gas: collision number, collision diameter and mean free path, behaviour of real gases: van der Waals equation and its applicability, the critical state: critical constants, law of corresponding states.	10
4. Liquid State: Surface tension, capillary action, measurement of surface tension, viscosity of liquids and its measurement, temperature effect on surface tension and viscosity	4
5. Solid State: Crystal lattices, space lattice, unit cell, crystal systems, law of rational indices, Miller indices, Bragg's equation, crystal structure of NaCl, graphite and diamond	4
6. Chemical Thermodynamics and its Applications: First law of Thermodynamics, Thermodynamic quantities (w, q, ΔU and ΔH) for isothermal and adiabatic reversible expansion of ideal gases, Kirchoff's relations, Joule effect, Joule-Thomson effect, Joule-Thomson coefficient of real (van der Waals) gases.	12
Second Law of Thermodynamics, Carnot cycle, concept of entropy and its physical significance, entropy changes for an ideal gas, entropy changes at constant pressure and volume, entropy change in spontaneous processes, free energy and maximum work functions, conditions for spontaneous changes and equilibrium, Gibbs-Helmholtz equation, Maxwell relations. Statement of third law of Thermodynamics.	
Outcomes: Students are expected to be able to understand the wave-particle duality, quantum numbers, and screening effect of electrons and its consequences. Students will be able to understand the basis of classification in the Modern Periodic Table and also will be able to explain the properties of various elements. The students will be able to understand the properties of solids, liquids, real and ideal gases. The students will learn various laws of thermodynamics.	
Suggested Books: 1. Basic Inorganic Chemistry, F. A Cotton, G. Wilkinson, and Paul L. Gaus, 3rd Edition (Indian Adaptation 2021), Wiley. 2. Concise Inorganic Chemistry, J. D. Lee, 5th Edition (1996), Chapman & Hall, London. 3. Principle of Physical Chemistry, B. R. Puri, L. R. Sharma and M. S. Pathania, Vishal Publishing Co., New Delhi. 4. Physical Chemistry, P. C. Rakshit, 5th Edition (1988), 4th Reprint (1997), Sarat Book House, Calcutta. 5. Physical Chemistry, P. W. Atkins, 8th Ed., Oxford University Press, New York. 6. Textbook of Physical Chemistry, K.L. Kapoor, vol. 1, Macmillan India Ltd, New Delhi. 7. Physical Chemistry, G. W. Castellan, 3rd Ed (1983), Addison-Wesley Publishing Company, Massachusetts.	

CHEMN12: Lab Work-I	
1. Redox titrations: (a) Redox Titrimetric Estimations Based on Iodometry (b) Redox Titrimetric Estimations with standard $K_2Cr_2O_7$ solution	
2. (a) Determination of Surface Tension of liquids. (b) Determination of Viscosity Coefficients of liquids.	
OR	
Inorganic/Organic Chemistry Laboratory: 1. Preparation of molecular models using wire and beads to study (a) Various hybridization geometries (b) Isomerism 2. Identification of Metal ions by borax bead test 3. Preparation of some simple and double salts. 4. Study of displacement of ions in aqueous solutions (e.g. Cu^{2+} by Zn) 5. Preparation of m-dinitrobenzene from nitrobenzene 6. Preparation of acetanilide from aniline	
Note: Experiments may be added/ modified subject to availability of facilities.	

CHEMN21: Basic Concepts of Chemistry-II	
Objectives: To understand the nature of chemical bonds and its various types using different approaches. To impart knowledge of general reactions mechanisms, preparation and reactivity patterns of alkyl halides, alcohols, ethers, and active methylene compounds.	
1. Chemical Bonding: i) A comparative account of the Valence Bond (VB) and Molecular Orbital (MO) approach of H_2 molecule. Molecular orbital treatment of Homonuclear diatomic molecules of elements of first and second period and Heteronuclear (CO, NO) diatomic molecules. Illustration of HOMO and LUMO. ii) Concept of Hybridization, Bent's Rule, VSEPR theory applied to the compounds of s and p block elements. iii) An introduction to multi-center bonding with illustrative examples of Li-alkyls and diborane. iv) Concept of multiple bonding ($p-p$ and $p-d$) with illustrative examples of BF_3 , P_4O_{10} , $(Me_3Si)_3N$ and SiF_4	6
2. Ionic Bonding: A brief introduction to Crystal Packing, Radius Ratio rule, and its exceptions Structures of a few representative crystal types (Rock salt, Zinc blende, Wurtzite, Rutile, Brookite and Anatase).	5
3. Metallic Bonding: Theories of bonding in metal: Valence Bond, Free electron, and Band theories	2
4. Weak Interactions: Brief introduction to Hydrogen bonding and other weak interactions: Effects on physical and chemical properties.	2
5. Electronic and Field Effect: Electronic (electromeric, inductive, resonance, mesomeric effect, and hyperconjugation) and steric effects on the acidity/basicity and reactivity of organic molecules.	
6. Stereochemistry: Fischer, Saw-horse and Newman projection formulae. Chirality-optical activity, enantiomerism and diastereoisomerism involving one and two chiral centers. Configuration: D/L, erythrose/threose, R/S nomenclatures; geometrical isomerism - E/Z nomenclatures. Conformations of n-butane.	6
7. Hydrocarbons: i. Alkenes: Addition reactions (electrophilic and free radical), hydration, hydroxylation, hydroboration, epoxidation, ozonolysis and polymerization. Addition reactions of conjugated dienes and Diels-Alder reaction. ii. Alkynes: Reduction, electrophilic addition, acidity and metal acetylides.	4
8. Alkyl Halides: Nucleophilic substitution: SN_1 , SN_2 and SN_i , Eliminations reactions: E_1 , E_2 , and E_{1cB} mechanisms. Elimination versus substitution reactions. Preparation and synthetic applications of Grignard reagents.	
9. Alcohols and Ethers: Preparation and reactions of primary, secondary, and tertiary alcohols and ethers.	4

10. Active methylene compounds: Preparation and synthetic applications of ethyl acetoacetate and diethyl malonate. (L)	5
Outcomes: The students will be able to understand how various types of chemical bonds are formed. Students are expected to be able to predict an organic compound's behavior. Students will be able to identify the mechanism of a specific chemical reaction. The goal of the unit is to provide an understanding of practically all facets of general organic chemistry and to provide a strong foundation in this area. The course aims to give a solid foundation of Alkyl/Aryl Halides, Alcohols, Phenols and Nitrogen-containing compounds and a grasp of nearly all its components.	
Suggested Books: 1. Basic Inorganic Chemistry, F. A Cotton, G. Wilkinson, and Paul L. Gaus, 3rd Edition (Indian Adaptation 2021), Wiley. 2. Concise Inorganic Chemistry, J. D. Lee, 5th Edition (1996), Chapman & Hall, London. 3. A Guide Book to Mechanism in Organic Chemistry, Peter Sykes, 6th Edition (1997), Orient Longman Ltd., New Delhi. 4. Stereochemistry of Organic Compounds, D. Nasipuri, 2nd Edition (1994), Wiley Eastern Ltd., New Delhi 5. Organic Chemistry, I. L. Finar, [Vol. I, 6th Edition (1973), Reprinted in 1980 & Vol. II, 5th Edition (1975), Reprinted in 1996], ELBS and Longman Ltd., New Delhi. 6. Organic Chemistry", 9th Edition, L. G. Wade, J. W. Simek & M. S. Singh, Pearson India Education Services Pvt. Ltd. 2020 (ISBN 978-93-893-4267-3). 7. Organic Chemistry, R. T. Morrison and R. N. Boyd, 6th Edition (1992), Prentice- Hall of India (P) Ltd., New Delhi. 8. Organic Chemistry, Paula Y. Bruice, 2nd Edition, (1998) Prentice-Hall, International Edition.	

CHEMN22: Lab Work-II	
1. Redox titrations: (a) Redox Titrimetric Estimations Based on Iodometry (b) Redox Titrimetric Estimations with standard $K_2Cr_2O_7$ solution	
2. (a) Determination of Surface Tension of liquids. (b) Determination of Viscosity Coefficients of liquids.	
OR	
Inorganic/Organic Chemistry Laboratory: 1. Preparation of molecular models using wire and beads to study (a) Various hybridization geometries (b) Isomerism 2. Identification of Metal ions by borax bead test 3. Preparation of some simple and double salts. 4. Study of displacement of ions in aqueous solutions (e.g. Cu^{2+} by Zn) 5. Preparation of m-dinitrobenzene from nitrobenzene 6. Preparation of acetanilide from aniline	
Note: Experiments may be added/ modified subject to availability of facilities.	

CHEMV31: Industrial Chemistry -I	
Objectives: To provide a comprehensive introduction to the science, formulation, production, and applications of cosmetics, soaps, and detergents.	
1. Introduction to Cosmetics, Soaps, and Detergents: Definition and classification of cosmetics, soaps, and detergents, Historical development and cultural significance, Regulatory framework and quality standards	4
2. Cosmetics: Types of cosmetics (skincare, makeup, haircare, etc.), Cosmetic ingredients and their functions (emollients, emulsifiers, preservatives, etc.), Formulation and manufacturing processes, Packaging and labelling requirements.	8
3. Soaps: Saponification process and soap chemistry, Types of soaps (bar, liquid, transparent, etc.), Specialty soaps (medicated, herbal, etc.), Soap additives and fragrances.	8
4. Detergents: Types of detergents (household, industrial, personal care, etc.), Surfactants and their role in detergency, Formulation and production processes, Environmental considerations and biodegradability.	8
5. Safety, Regulation and Consumer Awareness: Safety considerations in cosmetic and detergent production, Regulatory agencies and compliance requirements, Consumer awareness and product labelling.	2
Outcome: Students will gain practical knowledge about the ingredients, manufacturing processes, and quality control procedures involved in the production of these essential personal care products.	
Books Recommended: 1. Introduction to Cosmetic Formulation and Technology by Gabriella Baki, Kenneth S. Alexander 2. Soap Manufacturing Technology by Luis Spitz 3. Harry's Cosmeticology by Meyer R. Rosen, Richard L. Gabriele 4. Cosmetic and Toiletry Formulations by Ernest W. Flick 5. Handbook of Detergents, Part F: Production" by Uri Zoller 6. Soap Technology for the 1990s by H. P. Stevens 7. Detergent Science: A Primer on the Properties and Performance of Everyday Detergents" by Michael S. Showell 8. Handbook of Detergents, Part A: Properties by Uri Zoller 9. Cosmetic Safety Assessment: A Guide for Cosmetic Scientists, Regulators, Toxicologists and Dermatologists" by P. Elsner, H. I. Maibach	

CHEMV32: Industrial Chemistry Lab Work –I	
Preparation of Soaps, Detergents, and various types of cosmetics.	

CHEMN41: Physical Chemistry-II
Objectives: To impart knowledge on the application of classical thermodynamics on solution, surface and photochemistry and nuclear chemistry
1. Thermodynamics of Solutions: Partial molal quantities, chemical potential of a mixture of ideal gases, Gibbs-Duhem equation, chemical potential of real gases and fugacity, activity and activity coefficient, variation of fugacity with temperature and pressure, Lewis-Randall rule, thermodynamic functions of mixing (ΔG_{mix} , ΔS_{mix} , ΔV_{mix} , ΔH_{mix}), ideal solutions and their characteristic properties, Duhem-Margules equation and its applications, Henry and Raoult's laws, thermodynamics of colligative properties, relative lowering of vapour pressure, freezing point depression, elevation of boiling point and osmotic pressure, van't Hoff equation, measurement of osmotic pressure and determination of molecular weight of macromolecules.
2. Surface and Photochemistry: Langmuir adsorption isotherm, Multi-layer adsorption-BET equation (no derivation) and its application to surface area measurements; Gibbs Adsorption isotherm, surface reactions: kinetics of unimolecular surface reactions, inhibition and activation energy, nature of surface. Enzyme kinetics (Michaelis-Menten equation), Photochemical reactions: quantum efficiency, reasons for low and high quantum efficiency, kinetics of photochemical reactions: decomposition of HI, H_2 -Br $_2$ reaction, photostationary state: dimerisation of anthracene, Stern-Volmer relation.
3. Nuclear Chemistry: Radioactivity: general characteristics of radioactive decay kinetics, artificial radioactivity, detection and measurement of radioactivity: GM counter. Nuclear Reactions: Types of nuclear reactions, conservation of linear momentum and mass- energy in nuclear reactions, nuclear reaction cross section. Compound nucleus theory and its experimental verification. Radio chemistry: Probing by isotopes, preparation of radioisotopes, radiochemical principles in use of tracers, some typical applications: chemical, physicochemical and analytical applications, age determination. Radiation Chemistry: Interaction of nuclear radiations with matter- charged particle, neutrons and gamma rays. Radiolysis of water and aqueous solutions, radiation dosimetry.
Outcomes: After the completion of the course, the students should be able to 1. Apply the concept of chemical potential to explain the Gibbs free energy change due to mixing and calculate partial molar quantities. 2. Understand the concept of fugacity, activity, ideal and nonideal solution. 3. Explain the thermodynamic basis of colligative properties and applications 4. Learn about surface phenomenon, adsorption isotherms, BET Equation, the laws of absorption of light energy by molecules and the subsequent photochemical reactions 5. Understand the basics of nuclear chemistry
Suggested Books: 1. Physical Chemistry, P. C. Rakshit, 5th Edition (1988), Sarat Book House, Calcutta. 2. Textbook of Physical Chemistry, K.L. Kapoor, vol. 5th (2005), Macmillan India Ltd, New Delhi. 3. Essentials of Nuclear Chemistry, H.J. Arnikar, 4th Edition (2003), New Age International Publishers, New Delhi. 4. Physical Chemistry, I. Levine, 5th Ed., (2007), Tata McGraw Hill New Delhi. 5. Physical Chemistry, P. W. Atkins, 8th Ed., Oxford University Press, New York.

CHEMN42: Techniques in Chemistry	
Objectives: To introduce basic concepts about stoichiometry, acid-base equilibria, and separation methods.	
1. Introduction to Separation and Instrumental Methods Distillation, Crystallization, Precipitation, Extraction, Chromatography, Introductory ideas of UV-visible spectroscopy, Infrared Spectroscopy, and NMR Spectroscopy.	8
2. General Introduction to Analysis, and Stoichiometric Calculations Qualitative and quantitative analysis, primary standard, secondary standard, preparation of standard solutions, molarity, normality, molality, weight/weight percentage, weight/volume percentage, and ppm.	4
3. Concepts of Acid-Base Equilibria and Titrations General concepts of acids and bases, strength of conjugate acid/base pair, acid-base equilibria in water, pH calculations for strong acids and strong bases, concepts of acid base titration, complexometric titration, redox titration, and precipitation titration, indicators.	3
Outcomes: The student will understand about qualitative and quantitative analysis, titrations, and instrumentations on the separation techniques.	
Suggested Books: 1. Daniel C. Harris, <i>Quantitative Chemical Analysis</i> , 9 th Edition (2015), W. H. Freeman and Company, New York. 2. <i>Basic Concepts of Analytical Chemistry</i> , S.M. Khopkar, 2 nd edition (1998), New Age International Publications, New Delhi. 3. <i>Analytical Chemistry</i> , G.D. Christian, (2001) John Wiley & sons, New York	

CHEMV51: Food Adulteration & Detection	
Objectives: To provide students with a comprehensive understanding of food adulteration, its detection methods, and the principles of food quality assessment.	
1. Introduction to Food Adulteration: Definition and types of food adulteration, Historical perspective and global prevalence, Impact of food adulteration on public health and economy.	2
2. Common Adulterants in Food: Adulteration in staple foods (e.g., grains, pulses), Adulteration in dairy products, Adulteration in edible oils and fats, Adulteration in spices and condiments.	6
3. Methods of Food Adulteration Detection: Chemical methods (e.g., spectroscopy, chromatography), Physical methods (e.g., microscopy, texture analysis), Biological methods (e.g., DNA-based techniques).	12
4. Quality Assessment Parameters: Sensory evaluation, Nutritional analysis, Microbiological quality, Physical and chemical attributes.	4
5. Food Safety Regulations and Standards: International and national regulatory bodies (e.g., FDA, FSSAI), Food safety standards (e.g., ISO, HACCP), Compliance and enforcement mechanisms.	4
6. Case Studies and Real-world Applications: Notable incidents of food adulteration, Success stories in quality assurance and adulteration prevention.	2
Outcome: Students will learn about various types of food adulteration, its impact on public health, and techniques to ensure food safety and quality. The course will also cover relevant regulatory frameworks and standards.	
Books Recommended: 1. Food Adulteration and Quality Control by S. Suzumura, J. Yoshimura, and T. Kuwata 2. Food Adulteration and Analysis by Yashwant Pathak and Taylor Wallace 3. Food Analysis by S. S. Nielsen 4. Food Quality Assurance: Principles and Practices by Intez Ali	

CHEMV52: Food Adulteration & Detection Lab work	
Qualitative/quantitative analysis of food samples for moisture content, carbohydrates, proteins, fats, ash etc. Characterization of various adulterants. Identification of adulterants in real samples	

CHEMV61: Industrial Chemistry -II	
Objectives: To provide a comprehensive introduction to the science and technology of pigments, dyes, and paints.	
1. Introduction to Pigments, Dyes, and Paints: Definition, Historical development and classification and their significance. Chemical structures and properties.	3
2. Pigments: Inorganic vs. Organic pigments, Pigment synthesis and manufacturing processes, Colorimetry and color spaces, Applications in paints, plastics, printing inks, and more.	8
3. Dyes: Classification of dyes (acid, basic, direct, etc.), Dyeing processes (fiber, yarn, fabric, etc.), Dye synthesis and application techniques, Environmental considerations and regulations.	8
4. Paints: Types of paints (emulsion, enamel, lacquer, etc.), Formulation and composition of paints, Paint application methods (brushing, spraying, etc.), Surface preparation and finishing techniques.	6
5. Environmental impacts and sustainability: Environmental considerations in pigment, dye, and paint production, Waste management and pollution control, Green alternatives and sustainable practices.	5
Outcome: Students will gain a thorough understanding of the properties, synthesis, applications, and environmental impact of these important materials. Practical laboratory work will complement theoretical knowledge, enabling students to apply principles learned in lectures.	
Books Recommended: 1. Introduction to Pigments by E. Reinhold, R. M. Christie 2. Pigment Compendium; A Dictionary and Optical Microscopy of Historical Pigments" by Nicholas Eastaugh, Valentine Walsh, Tracey Chaplin, Ruth Siddall 3. The Chemistry of Synthetic Dyes by K. Venkataraman 4. Paint and Surface Coatings: Theory and Practice" by R. Lambourne, T. A. Strivens 5. Environmental Impact of Textile Dyeing" by Subramanian Senthilkannan Muthu	

CHEMV62: Industrial Chemistry Lab work -II	
Isolation of Pigments from plants Synthesis of Dyes Preparation of a few Paint samples	

CHEMN71: Analytical Chemistry-III	
Objectives: To introduce conceptual knowledge of advanced statistical methods, chemical equilibria, and stoichiometry.	
1. Analytical chemometrics: Propagation of measurement uncertainties (inaccuracy and imprecision). Useful statistical test: test of significance, the F test, the student 't' test, the chi-test, the correlation coefficient, confidence limit of the mean, comparison of two standard values, comparison of standard deviation with average deviation, comparison of mean with true values, significant figures, regression analysis (least square method for linear and non-linear plots), statistics of sampling and detection limit evaluation. Chemometrics for modeling and parameter estimation, factor analysis, resolution and pattern recognition.	15
2. Treatment of Equilibria: Solvents and solutions, general treatment of equilibria in aqueous medium involving monoprotic weak acid and weak base, and salts of weak acids and weak bases. Activity and concentration, Effect of electrolytes on chemical equilibria, Calculation of pH, Constructing titration curves from charge balance and mass balance equations, Acid-base titrations and theory of pH indicators, Complexation equilibria and complexometric titrations, Redox equilibria and redox titration, Theory of redox indicators, Precipitation reaction and precipitation titrations and theory of adsorption indicators.	8
3. Spectrophotometric Determination of Stoichiometry of Complexes: 7 Job's method of continuous variation, mole ratio and slope ratio analysis, Advantages and limitations, typical examples.	7
Outcomes: The students will establish knowledge on chemometrics, theory of various titrations, and stoichiometric calculations using instrumental techniques.	
Suggested Books: 1. R. L. Pecsok, L. D. Shields, T. Cairns and L.C. Mc William, Modern Methods of Chemical Analysis, 2nd Edition (1976), John Wiley, New York. 2. G. D. Christian, Analytical Chemistry, 5th Edition (1994), John Wiley & Sons, New York. 3. D. A. Skoog, D.M. West, F.J. Holler, S.R. Crouch, Analytical Chemistry - An Introduction, 7 th Edition (2000), Saunders College Publishing, Philadelphia, London. 4. J. H. Kennedy, Analytical Chemistry: Principles, 2nd Edition (1990), Saunders Holt, London.	

CHEMN72: Inorganic Chemistry-V	
Objectives: 1. To train students about the illustration of symmetry in complexes and the fundamentals of character tables; and Orgel diagrams of metal complexes. 2. To teach the biological perspectives of important metal ions.	
1. Symmetry-based concepts of energy level diagrams of metal complexes: Symmetry elements and operations, determination of point group of a molecule, Group representations, Features of specific character tables in reference to Orgel diagrams, Symmetry-based concepts of Orgel energy level diagrams of metal complexes. Introduction to Tanabe-Sugano diagrams (d ² configuration)	15
2. Bioinorganic chemistry: Metal ions in biological processes. Biological role of iron. Oxygen transport with reference to hemoglobin and myoglobin. Synthetic model complexes for oxygen transport. Cytochromes and iron-sulfur proteins; Electron Transfer reactions. Photosystems, Role of metalloenzymes. Biological role of alkali metal ions, and their transport with special reference to Na ⁺ /K ⁺ ATPase. Vitamin B12. Nitrogen Fixation.	15
Outcomes: Students shall be able to 1. Learn the importance of symmetry-based concepts in the field of coordination chemistry. 2. Develop an interdisciplinary approach to metal chemistry with the biological world.	
Suggested Books: 1. Physical Methods in Inorganic Chemistry, R. S. Drago, International Edn. (1971), Affiliated East-West Press, New Delhi. 2. Chemical Applications of Group Theory, F. A. Cotton 3. Inorganic Chemistry, J.E. Huheey, E.A. Keiter and R.L. Keiter, 4th Edn. (1993), Addison-Wesley Pub. Co., New York. 4. Bioinorganic Chemistry, I. Bertini, H. B. Gray, S. J. Lippard, J.S. Valentine, Viva Books, P. Ltd. New Delhi, South Asian Ed., 1998 (Reprint 2004). 5. Principles of Bioinorganic Chemistry, S. J. Lippard, J. M. Berg, University Science Books,	

6.	California 1994. Bioinorganic Chemistry: Inorganic Elements in the Chemistry of Life, W. Kaim, B. Schwederski, John Wiley & Sons, England, 2004.	
CHEMN81: Organic Chemistry-VI		
Objectives: The mechanistic aspects of asymmetric synthesis, pericyclic reactions, and the formation of carbon-carbon bonds in contemporary organic synthesis.		
1. Asymmetric Synthesis: Enolization (Lithium and Boron enolate), Stereoselective Aldol condensation reaction (chiral enolate & achiral aldehyde and achiral enolate and chiral aldehyde). The chiral pool, chiral auxiliaries, chiral catalysts and reagents, and enantiomeric excess.		
2. Pericyclic Reactions: Woodward-Hoffmann rules and molecular orbitals; Electrocyclic reactions; Cycloaddition reactions, 1,3-Dipolar cycloaddition and cheletropic reactions; Sigmatropic rearrangements; Group transfer reactions; Ene reactions.		
3. Modern organic synthesis: Retrosynthesis, Umpolung, Domino reactions and functional group interconversion.		
4. Formation of Carbon-Carbon Bonds via Organometallic reagents: Organozinc, Organocopper, and Organoborane reagents, Palladium catalyzed coupling reactions (Heck, Sonogashira, and Suzuki).		
Outcomes: The students will be able to understand 1. The mechanistic details of asymmetric synthesis of some bioactive molecules. 2. C-C Bond formation employing pericyclic reactions and organometallic reagents. 3. Mechanistic aspects some functional group interconversion.		
Suggested Books: 1. Clayden, Greeves, Warren and Wothers, Organic Chemistry, Oxford University Press, 2001. 2. Advanced Organic Chemistry-Reactions and Mechanisms 2e”–Maya Shankar Singh & Keshav Raghuvanshi, Pearson Education Pvt. Ltd., 2021 (ISBN 978-93-544- 9078-1) 3. M.B. Smith & Jerry March, March’s Advanced Organic Chemistry, 5th Edition (2001), John Wiley & Sons, New York. 4. G. S. Zweifel and M. H. Nantz, Modern Organic Synthesis, (2007), Freeman and Company, New York. 5. S. Kumar, V. Kumar, and S. P. Singh, Pericyclic Reaction; A Mechanistic and Problem-Solving Approach, 1st Edition (2015), Elsevier Inc. 6. T. H. Lowry and K. S. Richardson, Mechanism and Theory in Organic Chemistry, 3rd Edition (1998), Addison – Wesley Longman Inc. (IS Edition). 7. I. Fleming, Pericyclic Reactions, Oxford University Press, Oxford (1999). 8. W. Carruthers and Iain Coldham Modern Methods of Organic Synthesis Cambridge University Press 9. Charles H. Depuy and Orville L. Chapman, Molecular reactions and photochemistry, New Delhi: Prentice Hall of India, 1988.		

CHEMN82: Phys. Chem-VI		
Objectives: To impart knowledge on different advanced electrochemistry topics (like electrochemical double layer, electrode kinetics and cyclic voltammetry), basic concepts of polymer chemistry and micelles and colloids.		
1. Advanced Electrochemistry: Metal/Electrolyte interface: OHP and IHP, potential profile across double layer region, potential difference across electrified interface; structure of the double layer: Helmholtz- Perrin, Gouy-Chapman, and Stern models. Butler-Volmer equation under near equilibrium and non-equilibrium conditions, exchange current density, Tafel plot, polarizable and non-polarizable interfaces. Semiconductor (SC)/electrolyte interface: Creation of space charge region, capacity of space-charge, Mott-Schottky plots for n-type and p-type semiconductors, determination of flat-band potential and donor/acceptor densities, application of SC/electrolyte interface in solar cells, Fuel Cell, Battery. Cyclic voltammetry (CV) technique basics and applications.		15

2. Polymer Chemistry: Introduction, Classification of Polymers & Intermolecular forces of Polymers. Different types of molecular weight of polymer, polydispersity Index, Determination of molecular weight of polymers. Thermodynamic Aspects of Polymerization. Mechanism of Living Radical Polymerizations: Nitroxide mediated polymerization (NMP), Metal-catalyzed Living Radical Polymerization, Reversible Addition-Fragmentation Chain Transfer (RAFT) Radical Polymerization. Coordination polymerization, Ring-opening polymerization. Thermal and mechanical properties of polymers: Glass transition temperature, melting temperature, Rheological properties, industrial polymers.	8
3. Micelles and Colloids: Surface active agents and their classification, micellization, hydrophobic interaction, critical micellar concentration (cmc), factors affecting cmc of surfactants, thermodynamics of micellization, phase separation and mass action models, reverse micelles, solubilization. Stability and properties of colloids. Zeta potential, Stern double layer, Tyndall effect.	7
Outcomes: After the completion of the course, the students should be able to explain the origin of electrical double layer, different models describing double layer and electrical double layer capacitance and their applications. 1. Understand the kinetics of electrode reaction, exchange current density, over potential, transference coefficient, Tafel plot. 2. Know definition of polymer, molecular weight, different type of polymerization techniques with emphasis of radical polymerization, their kinetics, thermal and mechanical properties of polymer. 3. Understand the fundamentals of micelles and colloids, CMC, stability and properties of colloids, Zeta potential, Stern double layer, Tyndall effect. Suggested Books: 1. <i>Modern Electrochemistry</i> , Vol. 2 A & B, J.O'M. Bockris and A. K. N.Reddy, Plenum Press, New York. 2. F. W. Billmeyer, Jr., <i>Text Book of Polymer Science</i> , 3 rd Edition (1984), Wiley Interscience, New York. 3. <i>Fundamentals of Polymerization</i> , B. M. Mandal, Illustrated edition (2012), World Scientific; Singapore. 4. <i>Micelles: Theoretical and Applied Aspects</i> , Y. Moroi, Plenum Press, New York	

CHEMN7R1 (Anal. Chem-III), CHEMN7R2 (Inorg. Chem-V), CHEMN8R1 (Org. Chem- VI) AND CHEMN8R2 (Phys. Chem-VI) ARE NOT MENTIONED IN THE ABOVE LIST

CHEMN7R1: Analytical Chemistry-III	
Objectives: To introduce conceptual knowledge of advanced statistical methods, chemical equilibria, and stoichiometry.	
1. Analytical chemometrics: Propagation of measurement uncertainties (inaccuracy and imprecision). Useful statistical test: test of significance, the F test, the student 't' test, the chi-test, the correlation coefficient, confidence limit of the mean, comparison of two standard values, comparison of standard deviation with average deviation, comparison of mean with true values, significant figures, regression analysis (least square method for linear and non-linear plots), statistics of sampling and detection limit evaluation. Chemometrics for modeling and parameter estimation, factor analysis, resolution and pattern recognition.	15
2. Treatment of Equilibria: Solvents and solutions, general treatment of equilibria in aqueous medium involving, Effect of electrolytes on chemical equilibria, Calculation of pH, Constructing titration curves from charge balance and mass balance equations, Acid-base titrations and theory of pH indicators, Complexation equilibria and complexometric titrations, Redox equilibria and redox titration, Theory of redox indicators, Precipitation reaction and precipitation titrations and theory of adsorption indicators.	8
3. Spectrophotometric Determination of Stoichiometry of Complexes: Job's method of continuous variation, mole ratio and slope ratio analysis, Advantages and limitations, typical examples.	7
Outcomes: The students will establish knowledge on chemometrics, theory of various titrations, and stoichiometric calculations using instrumental techniques.	
Suggested Books: 1. R. L. Pecsok, L. D. Shields, T. Cairns and L.C. Mc William, <i>Modern Methods of Chemical Analysis</i> , 2nd Edition (1976), John Wiley, New York.	

2.	G. D. Christian, Analytical Chemistry, 5th Edition (1994), John Wiley & Sons, New York.	
3.	D. A. Skoog, D.M. West, F.J. Holler, S.R. Crouch, Analytical Chemistry - An Introduction, 7 th Edition (2000), Saunders College Publishing, Philadelphia, London.	
4.	J. H. Kennedy, Analytical Chemistry: Principles, 2nd Edition (1990), Saunders Holt, London.	

CHEMN7R2: Inorganic Chemistry-V		
Objectives:		
1.	To train students about the illustration of symmetry in complexes and the fundamentals of character tables; and Orgel diagrams of metal complexes.	
2.	To teach the biological perspectives of important metal ions.	
1. Symmetry-based concepts of energy level diagrams of metal complexes:	Symmetry elements and operations, determination of point group of a molecule, Group representations, Features of specific character tables in reference to Orgel diagrams, Symmetry-based concepts of Orgel energy level diagrams of metal complexes. Introduction to Tanabe-Sugano diagrams (d ² configuration)	15
2. Bioinorganic chemistry:	Metal ions in biological processes. Biological role of iron. Oxygen transport with reference to hemoglobin and myoglobin. Synthetic model complexes for oxygen transport. Cytochromes and iron-sulfur proteins: Electron Transfer reactions. Photosystems, Role of metalloenzymes. Biological role of alkali metal ions, and their transport with special reference to Na ⁺ /K ⁺ ATPase. Vitamin B12. Nitrogen Fixation.	15
Outcomes:		
Students shall be able to		
1. Learn the importance of symmetry-based concepts in the field of coordination chemistry.		
2. Develop an interdisciplinary approach to metal chemistry with the biological world.		
Suggested Books:		
1.	Physical Methods in Inorganic Chemistry, R. S. Drago, International Edn. (1971), Affiliated East-West Press, New Delhi.	
2.	Chemical Applications of Group Theory, F. A. Cotton	
3.	Inorganic Chemistry, J.E. Huheey, E.A. Keiter and R.L. Keiter, 4th Edn. (1993), Addison-Wesley Pub. Co., New York.	
4.	Bioinorganic Chemistry, I. Bertini, H. B. Gray, S. J. Lippard, J.S. Valentine, Viva Books, P. Ltd. New Delhi, South Asian Ed., 1998 (Reprint 2004).	
5.	Principles of Bioinorganic Chemistry, S. J. Lippard, J. M. Berg, University Science Books, California 1994.	
6.	Bioinorganic Chemistry: Inorganic Elements in the Chemistry of Life, W. Kaim, B. Schwederski, John Wiley & Sons, England, 2004.	

CHEMN8R1: Organic Chemistry-VI		
Objectives:		
The mechanistic aspects of asymmetric synthesis, pericyclic reactions, and the formation of carbon-carbon bonds in contemporary organic synthesis.		
1. Asymmetric Synthesis:	Enolization (Lithium and Boron enolate), Stereoselective Aldol condensation reaction (chiral enolate & achiral aldehyde and achiral enolate and chiral aldehyde). The chiral pool, chiral auxiliaries, chiral catalysts and reagents, and enantiomeric excess.	
2. Pericyclic Reactions: Woodward-Hoffmann rules and molecular orbitals; Electrocyclic reactions;	Cycloaddition reactions, 1,3-Dipolar cycloaddition and cheletropic reactions; Sigmatropic rearrangements; Group transfer reactions: Ene reactions.	
3. Modern organic synthesis:	Retrosynthesis, Umpolung, Domino reactions and functional group interconversion.	
4. Formation of Carbon-Carbon Bonds via Organometallic reagents:	Organozinc, Organocopper, and Organoborane reagents, Palladium catalyzed coupling reactions (Heck, Sonogashira, and Suzuki).	
Outcomes:		
The students will be able to understand		
1. the mechanistic details of asymmetric synthesis of some bioactive molecules.		
2. C-C Bond formation employing pericyclic reactions and organometallic reagents. mechanistic aspects some functional group interconversion.		

Suggested Books:	
1.	Clayden, Greeves, Warren and Wothers, Organic Chemistry, Oxford University Press, 2001.
2.	Advanced Organic Chemistry-Reactions and Mechanisms 2e”–Maya Shankar Singh & Keshav Raghuvanshi, Pearson Education Pvt. Ltd., 2021 (ISBN 978-93-544- 9078-1)
3.	M.B. Smith & Jerry March, March’s Advanced Organic Chemistry, 5th Edition (2001), John Wiley & Sons, New York.
4.	G. S. Zweifel and M. H. Nantz, Modern Organic Synthesis, (2007), Freeman and Company, New York.
5.	S. Kumar, V. Kumar, and S. P. Singh, Pericyclic Reaction; A Mechanistic and Problem-Solving Approach, 1st Edition (2015), Elsevier Inc.
6.	T. H. Lowry and K. S. Richardson, Mechanism and Theory in Organic Chemistry, 3rd Edition (1998), Addison – Wesley Longman Inc. (IS Edition).
7.	I. Fleming, Pericyclic Reactions, Oxford University Press, Oxford (1999).
8.	W. Carruthers and Iain Coldham Modern Methods of Organic Synthesis Cambridge University Press
9.	Charles H. Depuy and Orville L. Chapman, Molecular reactions and photochemistry, New Delhi: Prentice Hall of India, 1988.

CHEMN8R2: Phys. Chem-VI	
Objectives: To impart knowledge on different advanced electrochemistry topics (like electrochemical double layer, electrode kinetics and cyclic voltammetry), basic concepts of polymer chemistry and micelles and colloids.	
1. Advanced Electrochemistry: Metal/Electrolyte interface: OHP and IHP, potential profile across double layer region, potential difference across electrified interface; structure of the double layer: Helmholtz- Perrin, Gouy-Chapman, and Stern models. Butler-Volmer equation under near equilibrium and non-equilibrium conditions, exchange current density, Tafel plot, polarizable and non-polarizable interfaces. Semiconductor (SC)/electrolyte interface: Creation of space charge region, capacity of space-charge, Mott-Schottky plots for n-type and p-type semiconductors, determination of flat-band potential and donor/acceptor densities, application of SC/electrolyte interface in solar cells, Fuel Cell, Battery. Cyclic voltammetry (CV) technique basics and applications.	15
2. Polymer Chemistry: Introduction, Classification of Polymers & Intermolecular forces of Polymers. Different types of molecular weight of polymer, polydispersity Index, Determination of molecular weight of polymers. Thermodynamic Aspects of Polymerization. Mechanism of Living Radical Polymerizations: Nitroxide mediated polymerization (NMP), Metal-catalyzed Living Radical Polymerization, Reversible Addition-Fragmentation Chain Transfer (RAFT) Radical Polymerization. Coordination polymerization, Ring-opening polymerization. Thermal and mechanical properties of polymers: Glass transition temperature, melting temperature, Rheological properties, industrial polymers.	8
3. Micelles and Colloids: Surface active agents and their classification, micellization, hydrophobic interaction, critical micellar concentration (cmc), factors affecting cmc of surfactants, thermodynamics of micellization, phase separation and mass action models, reverse micelles, solubilization. Stability and properties of colloids. Zeta potential, Stern double layer, Tyndall effect.	7
Outcomes: After the completion of the course, the students should be able to 1. Explain the origin of electrical double layer, different models describing double layer and electrical double layer capacitance and their applications. 2. Understand the kinetics of electrode reaction, exchange current density, overpotential, transference coefficient, Tafel plot. 3. Know definition of polymer, molecular weight, different type of polymerization techniques with emphasis of radical polymerization, their kinetics, thermal and mechanical properties of polymer. 4. Understand the fundamentals of micelles and colloids, CMC, stability and properties of colloids, Zeta potential, Stern double layer, Tyndall effect.	
Suggested Books: 1. Modern Electrochemistry, Vol. 2 A & B, J.O’M. Bockris and A. K. N. Reddy, Plenum Press, New York. 2. F. W. Bill mayer, Jr., Text Book of Polymer Science, 3rd Edition (1984), Wiley Inter science, New York.	

3.	Fundamentals of Polymerization, B. M. Mandal, Illustrated edition (2012), World Scientific; Singapore.	
4.	Micelles: Theoretical and Applied Aspects, Y. Moroi, Plenum Press, New York	

CHESE11: Quality Control in Chemical Analysis	
Objectives: To train the students with the quality control practices being adopted in the chemical and pharmaceutical industries. The course includes the principles related to statistical handling of data; quality control principles applicable in the laboratories as well as quality management systems including those laid down by the International Standard Organization (ISO).	
1. Statistical Quality Control Techniques: Statistical treatment of data, Control charts, Performance Evaluation uncertainties in measurement, Validation of analytical methods, Elements of quality Assurance, Principles of Good Laboratory Practices, Good manufacturing Practices in Drugs and Pharmaceutical Industries, Accreditation of QC laboratories, Tools and Mechanisms ICH Guidelines on Drug substances and Products.	25
2. Quality Management and Standards: Quality Management System (QMS), Quality management concepts and principles: ISO 9001:2000 QMS, Case studies on ISO 9001: 2000 in chemical industries. ISO 14000 Series of Standards. Total Quality Management (TQM) in Chemical Industry. Six Sigma Approach to Quality: Applying Six Sigma to chemical Industries.	20
Outcome: The students will be able to learn the quality control practices and also to apply them.	
Books Recommended: 1. Douglas M. Considine, Handbook of applied instrumentation, McGraw-Hill. 2. George T. Austin, Shreve's Chemical Process Industries, McGraw-Hill. 3. Willing, S.W., & Stoker, Good Manufacturing Practices for Pharmaceuticals, Marcel Dekker, New York. 4. Guarino, R.A., New Drug Approval Process, Marcel Dekker, New York.	

CHESE21: Instrumentation in Chemical Industry	
Objectives: To develop the skills of Chemical Analysis using sophisticated modern equipments to the students Using sophisticated Instruments in Chemical analysis and Characterization. This will enable them to understand the functioning of the equipment and their proper handling for qualitative and quantitative analysis of molecules and materials.	
1. Thermoanalytical methods: Thermoanalytical methods: thermogravimetry, factors affecting thermogravimetric curves, derivative thermogravimetry (DTG), thermobalance. Applications of thermogravimetry, differential thermal analysis (DTA), factors affecting DTA curves, instrumentation, and application of DTA. Differential scanning calorimetry (DSC), theory, instrumentation, applications of DSC, thermometric titrations, and principal classification.	15
2. Electrochemical Analysis: Potentiometry – fundamentals, reference electrodes (hydrogen, calomel, silver, and silver chloride electrode). Indicator and ion selective electrode, instrumentation and measurement of cell and emf-conductance and conductometric titrations.	15
3. Structure Analysis: Diffraction Techniques: Elementary theory (Neutron. Electron and X-ray diffraction), instrumentation, and applications of X-ray methods (emission, absorption, diffraction, and fluorescence methods). Spectroscopic Techniques: Atomic absorption, Spectroscopy, Atomic emission spectrometry, UV-visible molecular absorption spectrometry, Molecular luminescence spectroscopy (fluorescence, phosphorescence, chemiluminescence).	15
Outcome: The students will learn the theoretical background as well as practical handling of sophisticated instruments used for chemical analysis.	

Books Recommended:	
1.	H. Willard, L. Merrit, J.A. Dean & F.A. Settle, Instrumental Method of Analysis, 7th Ed. CBS (2006).
2.	G. D. Christian, Analytical Chemistry, 5th Ed., John Wiley & Sons, New York (2004).
3.	D.A. Skoog, D.M. West, F.J. Holler, S.R. Crouch, Analytical Chemistry: An introduction, 7th Ed., Saunders College Publishing, Philadelphia, London (2003).

CHESE31: Synthetic Methodologies	
Objectives: To stay abreast with the present-day chemical synthetic methods and strategies. Synthesis is one of the core topics of chemistry which requires knowledge and skill to handle air/moisture-sensitive compounds. At the same time, there are techniques to use non-aqueous solvents such as liquid ammonia at low temperatures.	
1. Purification and drying of solvents: Water and common non-aqueous solvents. Distillation (including fractional distillation and distillation under low pressure). Removal of hazardous impurities from solvents. Drying of non-aqueous solvents.	8
2. Synthesis: Preparation of sodium solution in liquid ammonia and its application. Synthesis and Applications of Grignard Reagent	15
3. Click Chemistry: Regioselectivity in Click Chemistry, CuAAC, RuAAC, Bioorthogonal Chemistry, Organocatalyzed Click, Sufex, Thiol-ene Click, and Their Applications.	15
4. Reduction using metal hydrides	7
Outcome: The students will learn the recent advances including some conventional (but tricky) methods of synthesis. They will be able to handle and perform reactions under an inert atmosphere.	
Books Recommended: 1. Practical Organic Chemistry, A.I. Vogel 2. Click Chemistry, V.K. Tiwari, M. K. Jaisawal, S. Rajkhowa, S.K. Singh, Springer, Singapore, 2024 3. IL Finar Vol. 1 4. E. S. Gould	

CHEMD11: Basic Principles of Chemistry and Its Applications in Daily Life	
Objectives: Chemistry is the central science. By studying chemistry one can find answers of many questions that arise to an inquisitive mind. This course is to be offered to students who are curious and interested to know the realities of the world in terms of atoms, molecules, materials and their chemical reactions. Further, some light will also be thrown on the chemistry of some materials that we come across in our everyday life.	
1. Basic Principles of Chemistry: Atoms, molecules, and their electronic structures. Types of Chemical bonding. Basic concept of Hybridization. Energetics of bond formation, Types of intermolecular/intramolecular forces, and molecular association for the formation of materials.	10
2. Chemical Reactions: Thermodynamic basis of chemical changes. Enthalpy, entropy, and free energy change during a chemical reaction. Types of Chemical Reactions. The concept of oxidation number, Reducing and oxidizing agents.	10
3. Applications of Chemistry in daily life : a) Dyes and pigments: General idea about the chemistry of dyes, Classification of dyes, Origin of the colors in the dyes and pigments, and application in garment and printing industries. b) Polymers: Natural and synthetic polymers, synthesis, properties, and applications. (Examples: polyester, rubber, nylon, bakelite, etc). Introduction to biopolymers. Application of synthetic polymers in surgical treatments and transplantations. c) Food and beverage industry: use of preservatives, food adulteration, flavor enhancers, artificial sweetening agents, and antioxidants. d) Fragrance: Basic principles, types of fragrances, and their applications. e) Cleansing agents: Soaps, detergents, Surfactants, and underlying chemistry of cleansing agents.	25

4. Reduction using metal hydrides	
Outcome: The learners of this course will find answers to several fundamental questions about molecules, materials and their chemical reactivity. They will also gain knowledge about several substances, such as polymers, pigment & dyes, fragrance, cleansing agents etc. that are commonly used in our daily life.	
Books Recommended: 1. Basic Inorganic Chemistry, F. A Cotton, G. Wilkinson, and Paul L. Gaus, 3rd Edition (1995), John Wiley & Sons, New York. 2. Concise Inorganic Chemistry, J. D. Lee, 5th Edition (1996), Chapman & Hall, London. 3. Principle of Physical Chemistry, B. R. Puri, L. R. Sharma and M. S. Pathania, Vishal Publishing Co., New Delhi. 4. Physical Chemistry, P. W. Atkins, 8th Ed., Oxford University Press, New York. 5. Textbook of Physical Chemistry, K.L. Kapoor, vol. 1, Macmillan India Ltd, New Delhi. 6. Hall, A.J. (Eighth Edition) The Standard Hand Book of Textiles Butter Worth. London. 7. Clark, W. An Introduction to Textile Printing, Butter Worth, London, 3rd Edition, 1971. 8. The Chemistry and Application of Dyes - D. R. Waring and G. Hallas, Springer, 1990. 9. Introduction to polymer chemistry, R. Seymour, Wiley –Interscience (1981) 10. Darrell D. Ebbing and Mark S. Wrihton, 5th Edition 1998, A.I.T.B.S. Publishers & Distributors (Regd.) J-5/6 Krishna Nagar, Delhi-110051 (INDIA) 11. Pearson D; “Chemical Analysis of Food”, Chemical Publ. Co. New York, 1971. 12. Fragrance chemistry: The Science of the sense of smell, edited by Ernst T. Theimer, Rumson, New Jersey, Academic Press, 1982, New York London.	
CHEMD21: Energy & Metallurgy	
Objectives: Chemistry is the central science. By studying chemistry one can find answers of many questions that arise to an inquisitive mind. This course is to be offered to the students who are curious and interested to know realities of the world in terms of energy and its utilization. How solar energy is used by the plants and how we can manage various energy sources. Further, some light will also be shed on the importance of metals and chemistry of metal extraction.	
1. Photosynthesis: Introduction- a key process for life on earth, Light reaction, Dark reaction, Factors affecting photosynthesis, Photorespiration, artificial photosynthesis. 2. Energy Sources and storage: Introduction to renewable (green) and non-renewable sources of energy. Requirement and consumption of energy in India. Worldwide potential of these sources.	8
Understanding (i) Definition (ii) source (iii) advantages/disadvantages (iv) environmental aspects (v) future scope/sustainability along with specific attributes of each of the following types: Non-Renewable: a) Thermal-Oil, natural gas, and coal Oil - Origin, composition, specification, and characteristics of petroleum products Natural gas - Classification (natural and synthetic gas), Natural gas reserves - India and World Coal: Coal reserves- India and World, Origin, composition, Action of heat on coal, Coal derived chemicals Nuclear energy - Radioactivity, Different types of radiation, Nuclear fission, Basic components of Nuclear power reactor. Renewable: a) Solar energy - Importance, storage, applications (solar cooker, solar weather heater, solar greenhouses, solar cell, absorption air conditioning) b) Biomass energy - biomass conversion, biogas generation (types of plants needed, plant wastes, materials), Utilisation of biogas, energy farming (biomass production for energy) c) FuelCell Li-ion battery	27
3. Metals and Metallurgy: Role of metals in our life with special mention of Iron and Copper, Recovery of Iron and copper from its ores, Stainless Steel, corrosion of metals, and its prevention.	10
Outcome: The modern life is highly dependent on availability of energy. With limited resources of fossil fuels need for alternative sources have become evident. The students will be able to learn the present scenario about the conventional energy sources, their limitations and problems. The students will be made aware of the alternative sources of energy which have already been part of our life to some extent and are going to be very important in coming years.	
1. G.W. Vanloon, S.J. Duffer, Environmental Chemistry- A Global Perspective, Oxford University Press (2000). 2. A.K. De, Environmental Chemistry, 4th Edition (2000), New Age International Private Ltd., New Delhi. 3. Non-conventional energy sources - G.D. Rai - Khanna Publishers, New Delhi	

4.	Godfrey Boyle, "Renewable Energy, Power for a sustainable future", 2004, Oxford University Press, in association with The Open University.
5.	Vladimir Strezov, Hossain Md. Anawar, Renewable Energy Systems From Biomass: Efficiency, Innovation and Sustainability, 1st Edition.
6.	S.P. Sharma & Chander Mohan, "Fuels & Combustion", Tata McGraw Hill Publishing Co. Ltd., 1984
7.	Sarkar S. (2010); Fuels and Combustion, Third Edition, CRC Press
8.	"Photosynthesis" - J. J. Eaton-Rye, B. C. Tripathy, T. D. Sharkey, Springer, 2012.
9.	Colin Baird, Environmental Chemistry, W.H. Freeman and Company, New York (1995).

CHEMD31: Environmental Chemistry	
Objectives: Sustainable development is one of the most important challenges to the human civilization today. Indiscriminate use of natural resources, such as fossil fuels has polluted not only the air, soil and water bodies near to us but also has damaged the outer atmosphere. This course will throw light on various aspects of environmental chemistry, Pollution and its causes and also potential remedies with an idea of sustainable development.	
1. Introduction: Concept and scope of environmental chemistry, Pollution versus contamination, Receptor, Sink, and pathways of a pollutant. Terminology (DO, BOD, COD, TLV) used in environmental analysis and their estimation.	7
2. Environmental Segments: Atmosphere- Regions of atmosphere, Carbon Footprint, Greenhouse effect, and Global warming, ozone hole. Hydrosphere- Microorganisms in aquatic chemical reactions, eutrophication versus oligotrophication. Lithosphere- Soil profile, soil properties, organic and inorganic components of soil, micro- and macro-nutrients Biosphere.	22
3. Natural cycles of the environment: Hydrological cycle, oxygen cycle, nitrogen cycle, phosphate cycle, and sulfur cycle.	6
4. Pollution: Sources of air pollution, Auto exhaust emission, and its prevention. Air quality parameters, EL Nino phenomenon, Water pollution, Acid rains, Waste-water treatment: domestic and industrial effluents. Water quality parameters. Recent environmental concerns in the national and global context.	10
Outcome: After studying this course the students will be able to understand the various causes of pollution, how these can be minimized. How can scientific and industrial development go ahead without disturbing the natural balance so that not only human beings but also other flora and fauna can survive on this earth?	
Text Books: 1. Environmental Chemistry A.K. De, 3rd edition (1994), Willey Eastern, New Delhi. 2. Collin Baird, Environmental Chemistry, 4th edition (2000), New age international Private Ltd., New Delhi. 3. G. W. Vanloon, S. J. Duffer, Environmental Chemistry – A Global Perspective, Oxford University Press (2000).	